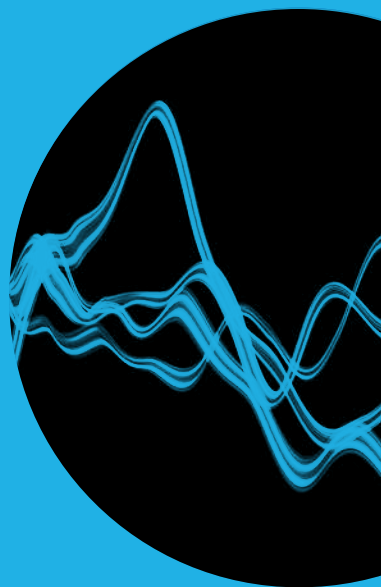


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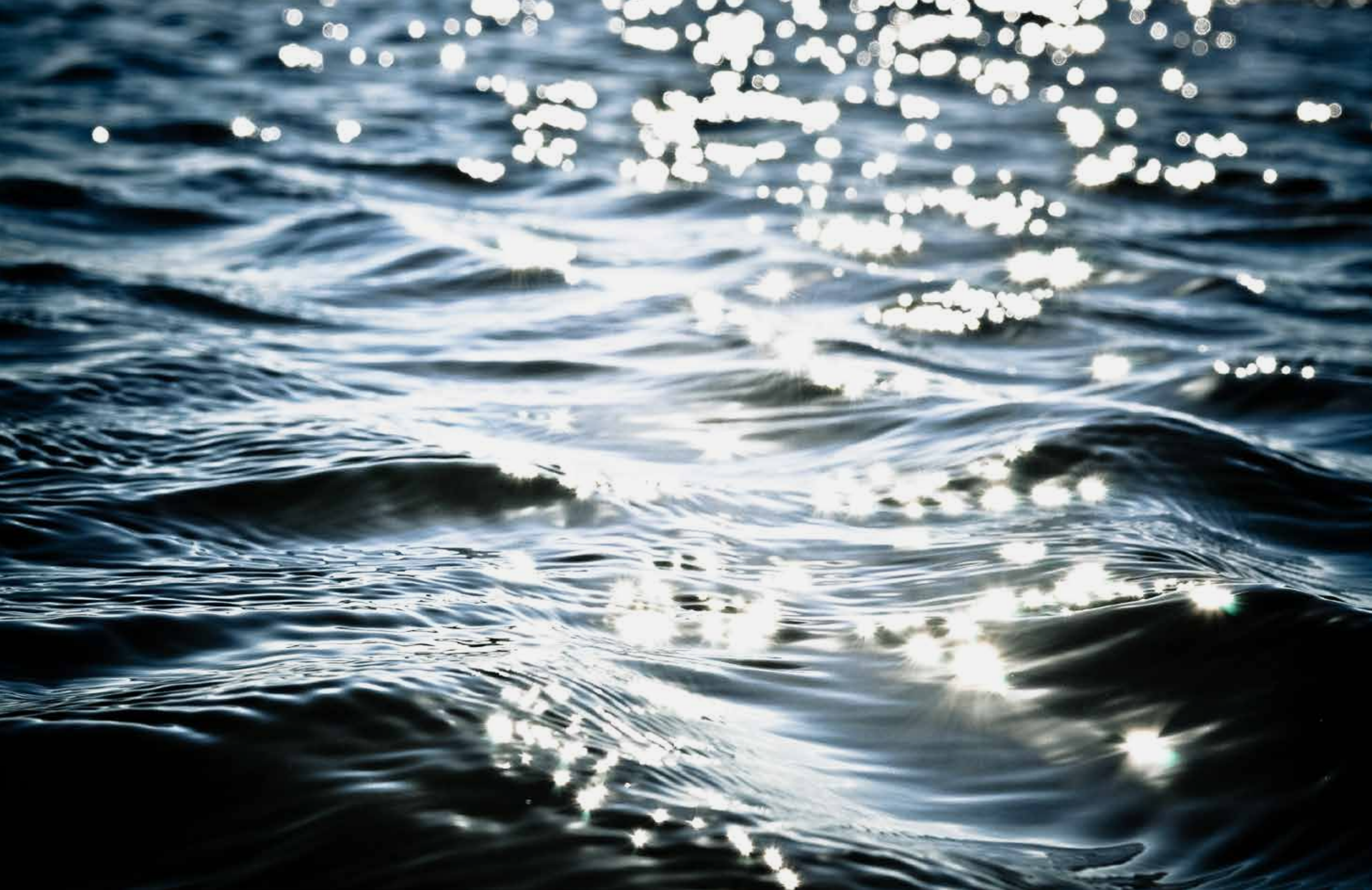


FUTURE NETWORK
TRENDS
FOR METAVERSE
MOBILITY

ENERGY
OPTIMIZED
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MODERNIZATION

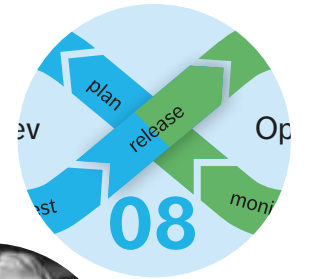
CREATING ZERO-TOUCH
AUTONOMOUS
NETWORKS





08 5G ARCHITECTURE FOR HYBRID AND MULTI-CLOUD ENVIRONMENTS

To maximize the flexibility of communication service providers to deploy 5G solutions in the cloud environments of their choice – public, private or both – cloud-native applications must be designed for hybrid and multi-cloud deployments.

**20 CREATING AUTONOMOUS NETWORKS WITH INTENT-BASED CLOSED LOOPS**

Intent-based operation is a new paradigm for telecommunication systems that is essential to the creation of the zero-touch, autonomous networks of the future. Our proofs of concept reveal that the technologies needed to create highly capable autonomous networks are already available today.

**30 HOLOGRAPHIC COMMUNICATION IN 5G NETWORKS**

The emergence of lightweight augmented reality (AR) glasses and powerful 3D compression algorithms make it possible to deploy AR use cases using existing 5G technology, which represents a significant step toward the Internet of Senses.

**40 FEATURE ARTICLE*****Future network trends driving universal metaverse mobility***

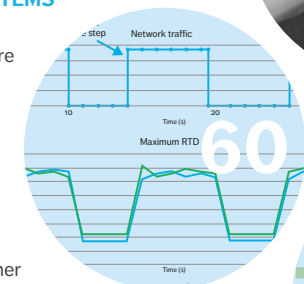
Ericsson CTO Erik Ekudden explores the three technology trends that are transforming telecom networks into a unified innovation platform that will enable universal metaverse mobility.

**50 ENERGY-OPTIMIZED NETWORK MODERNIZATION**

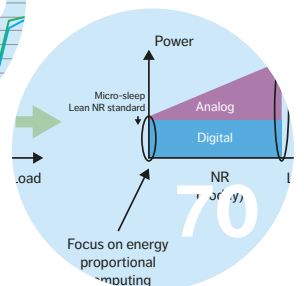
Building a more resilient, robust and cost-efficient radio access network requires a transition toward a smart energy setup at information and communications technology (ICT) sites. Ericsson's innovative approach helps ICT site owners enable demand-and-response energy services within the ICT network that will reduce the energy bill to net zero.

**60 ENERGY-EFFICIENT PACKET PROCESSING IN 5G MOBILE SYSTEMS**

Communication service providers can significantly reduce the amount of energy consumed by their data centers through a combination of hardware offload, frequency scaling and the application of micro-sleeps in packet processing nodes.

**70 IMPROVING ENERGY PERFORMANCE IN 5G NETWORKS AND BEYOND**

The lean design of New Radio has led to unprecedentedly low energy consumption in live 5G networks. To enable novel deployment solutions and the use of smaller, lighter products, 6G standards should include further reductions in fixed idle-mode energy usage and peak power requirements.



Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

ADDRESS

Ericsson
SE -164 83 Stockholm, Sweden
Phone: +46 8 7190000

PUBLISHING

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PUBLISHER

Erik Ekudden

EDITOR

Tanis Bestland (Nordic Morning)

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Future network trends driving
universal metaverse mobility
by Erik Ekudden

ART DIRECTOR

Carola Pilarz (Nordic Morning)

PROJECT MANAGER

Susanna O'Grady (Nordic Morning)

LAYOUT

Carola Pilarz (Nordic Morning)

ILLUSTRATIONS

Jenny Andersén (Nordic Morning)

SUBEDITORS

Ian Nicholson (Nordic Morning)
Paul Eade (Nordic Morning)

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CREATING A UNIFIED NETWORK PLATFORM FOR INNOVATION

■ **THE OFTEN SURREAL** experience of living through a global pandemic has profoundly changed the way that many organizations – and the individuals that comprise them – think about telecommunication networks. There is now a widespread, shared understanding of the critical role that networks play in almost every aspect of business and society. The desire for greater digitalization to enhance social, economic and environmental outcomes for the benefit of all has never been stronger.

In light of this, it's no surprise that expectations about future critical network capabilities and performance have grown considerably among stakeholders across business, societal and governmental spheres in a very short period of time. Our industry is well prepared to meet these expectations, thanks to our long-term, continuous investments in 5G evolution and in 6G research, which have long been aimed at creating a universal connectivity platform to support all manner of innovation, automation and digitalization.

In my 2022 technology trends article, I explore the most important future network trends that are driving the development of both 5G and 6G. There are three of them this year: (1) a unified network platform that enables business-driven enterprise innovation, (2) connectivity for an immersive user experience, and (3)

GREATER DIGITALIZATION TO ENHANCE SOCIAL, ECONOMIC AND ENVIRONMENTAL OUTCOMES FOR THE BENEFIT OF ALL

the quest for high-performing networks. To learn more about this year's trends and their implications, check out the article on page 40.

This issue of the magazine also includes six other articles that highlight progress in important research areas including hybrid and multi-cloud deployments, self-adapting automation and energy efficiency. Energy efficiency is a particularly hot topic right now for obvious reasons, which is why I am pleased to be able to include three articles on the subject in this issue. The network energy performance article on page 70 may be the most logical place to start. The authors begin by outlining all the ways in which the lean design of the NR standard enables communication service providers to significantly lower the energy consumption of their 5G networks in comparison to what was possible with 4G.

Anyone who is keen to learn more about the future of holographic communication will benefit from reading the article on page 30. The authors have explored the feasibility of using 5G networks to deliver holographic communication, which is recognized as one of the most anticipated XR use cases. Their end-to-end architecture for high-quality holographic communication makes it possible to both reduce the energy consumption of mobile devices and the

end-to-end latency by moving high performance computing to the 5G network.

We hope you enjoy this issue of our magazine and that you will pass it along to your colleagues and business partners. You can find both PDF and HTML versions of all the articles at:
www.ericsson.com/ericsson-technology-review



ERIK EKUDDEN
SENIOR VICE PRESIDENT,
CHIEF TECHNOLOGY OFFICER

5G architecture for hybrid and multi-cloud environments

A unified approach to developing, deploying and operating 5G services – including 5G RAN, Core, OSS and BSS applications – in public and private cloud environments is a key enabler for communication service providers to successfully adopt a hybrid and multi-cloud strategy. The main benefits are faster time to market and lower total cost of ownership.

ANTONIO ALONSO, HENRIK SAAVEDRA PERSSON, HOSSEIN KASSAEI

The amazing growth of cloud infrastructure that we see today is the result of many years of open-source innovation combined with the efforts of hyperscale cloud providers such as Amazon Web Services, Microsoft Azure and Google Cloud Platform, as well as large and medium-size communication service providers that have created their own private clouds in-house.

■ In addition to capacity, cloud infrastructure is growing geographically from national and regional levels all the way to edge and far edge locations, bringing the infrastructure to within a few milliseconds of where the developers want to deploy their applications, which is as close to the consumers as possible. 5G adds ultra-low latency, high capacity and programmable connectivity to such

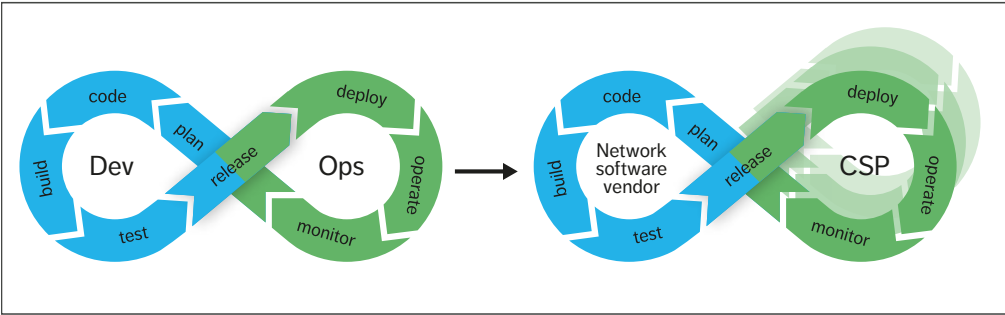


Figure 1 The DevOps and aaP business models

massive-scale infrastructure. The combination of ubiquitous infrastructure and superior connectivity provides application developers with a unique and unprecedented opportunity for innovation.

DevOps in the telecom context

DevOps – a set of practices that brings together software development and IT operations with the goal of shortening the development and delivery cycle and increasing software quality – is often thought of and discussed in the context of a single company or organization. The company usually develops the software, operates it and provides it as a service to customers, according to the software-as-a-service (SaaS) model. Within this context, it is easier to have full control over the entire flow, including full knowledge of the target deployment environment.

In the telecom space, by contrast, we typically follow the as-a-product (aaP) business model, in which software is developed by network software vendors such as Ericsson and provided to

communication service providers (CSPs) that deploy and operate it within their network. This business model requires the consideration of additional aspects.

As shown in Figure 1, the most important contrasts between the standard DevOps SaaS model and the telecom aaP model are the multiplicity of deployment environments and the fact the network software vendor development teams cannot know upfront exactly what the target environment looks like. Although a SaaS company is likely to deploy and manage its software on two or more different cloud environments, this is inevitable within telco, as each CSP creates and/or selects its own cloud infrastructure.

Implications of a hybrid and multi-cloud strategy

CSPs are expanding their cloud infrastructures through partnerships with hyperscale cloud providers (HCPs) and are increasingly adopting a hybrid and multi-cloud strategy. A recent Ericsson

Terms and abbreviations

aaP – as a Product | ACK – AWS Controllers for Kubernetes | API – Application Programming Interface | ASO – Azure Service Operator | AWS – Amazon Web Services | CNCF – Cloud Native Computing Foundation | CSP – Communication Service Provider | GCP – Google Cloud Platform | HCP – Hyperscale Cloud Provider | IaC – Infrastructure as Code | IAM – Identity and Access Management | SaaS – Software as a Service | TCO – Total Cost of Ownership | TTM – Time to Market

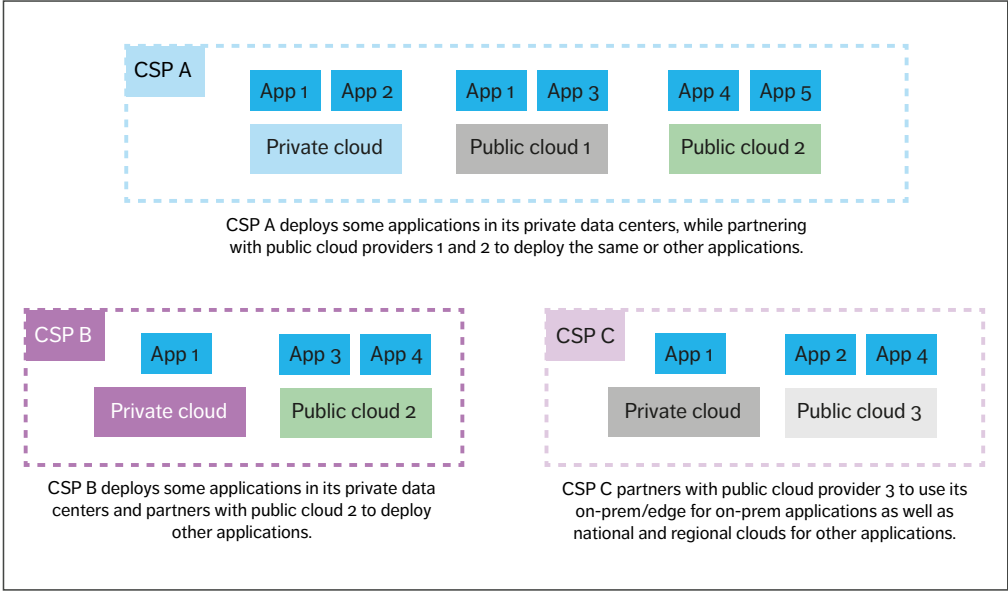


Figure 2 Examples of hybrid and multi-cloud deployment scenarios that applications must be able to support

Technology Review article presented two architecture scenarios for the deployment of telco workloads, showing how telco workloads can be deployed in a dual-stack architecture in a CSP's cloud infrastructure or deployed together with enterprise and consumer applications in an integrated-stack architecture on public clouds [1]. As Figure 2 illustrates, this approach could lead to a lot of diversity and heterogeneity in the deployment targets for network software vendors.

Kubernetes has established itself as the de facto orchestrator for cloud-native applications. As it is offered as a service by all major cloud providers, it can be viewed as a portability layer for application workloads. In addition, in the integrated-stack architecture, CSPs may also want to utilize other managed services that HCPs offer as part of their partnership. These managed services span across a wide range of capabilities from operational and analytics databases to observability and security services.

Aside from avoiding the cost of a DIY approach, one of the other main advantages of using an HCP service is the ability to delegate responsibility for aspects such as performance, availability and fault tolerance, updates and upgrades, security and scalability to the cloud provider. Another benefit is the ability to create alignment between different network software vendors in terms of data management, observability, security, and other non-functional requirements. However, designing and operating an application that is capable of utilizing such a diverse set of HCP managed services also creates several challenges for network software vendors and CSPs alike.

The main challenges to overcome in a hybrid and multi-cloud strategy are:

1. Maintaining portability
2. Controlling the total cost of ownership (TCO)
3. Optimizing productivity and time to market (TTM).

Portability

As network software vendors must have the flexibility to deploy 5G application workloads wherever CSPs decide it makes the most business sense to do so, it is essential that network software vendors support the CSPs' strategy of adopting a hybrid or multi-cloud approach and enable CSPs to avoid HCP vendor lock-in. HCPs continue to innovate and offer great managed services, but in many cases with different and proprietary application programming interfaces (APIs). One challenge to address therefore is to design portable applications in such a way that CSPs can use HCP managed services where desired, while allowing the decision on which managed service to use to be deferred to deployment time.

Total cost of ownership

Three important aspects influence TCO for CSPs: continuous verification and deployment, monitoring and operations, and security and compliance. In a hybrid and multi-cloud environment, these tasks tend to become more complex given the diversity of infrastructure environments and managed services. Unless these tasks are performed with extreme efficiency, the larger number of tracks to verify and increased operational complexities could lead to higher operating expenses, with a negative effect on TCO. Decisions about which managed service(s) to use and which services the CSP itself will deploy could also have a significant impact on TCO. These decisions depend on the use cases, usage patterns and the application's scale.

Optimizing productivity and time to market

If applications for hybrid and multi-cloud deployments are not developed correctly from the beginning, the long-term developer experience may be suboptimal. Take, for example, an application that is built with one cloud environment in mind that utilizes all the relevant managed services from that cloud. If it is later decided that this application will be deployed in a different cloud environment with a different set of managed services (or with similar services but different orchestration APIs), application developers will have to spend considerable time and effort migrating the application. Time and resources that could otherwise be spent on developing new and enhanced features will be spent on porting and adapting the application to a different environment, which reduces productivity and could result in slower TTM.

A unified approach to development, deployment, security and operations

A multi-cloud-native application is a cloud-native application that can be deployed in different clouds with minimal need for refactoring or redesign, while utilizing managed services offered in each cloud. As such, it must be designed and built with multiple clouds in mind from the beginning. As shown in Figure 3, this necessitates a holistic approach to the entire application life cycle from design and development to verification, deployment and operations. To the greatest extent possible, this approach also aims to unify all these tasks across multiple cloud environments.

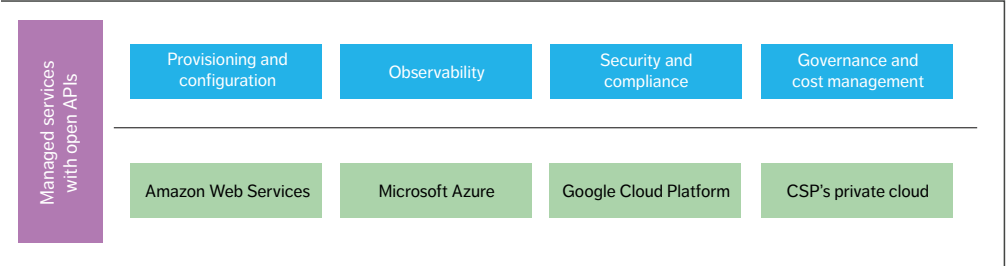


Figure 3 Key enablers for a multi-cloud-native application

MAJOR CLOUD PROVIDERS ARE TAKING NOTICE OF THE POPULAR OPEN-SOURCE PROJECTS WITHIN THE CNCF

Development based on open APIs

The major cloud providers are taking notice of the popular open-source projects within the Cloud Native Computing Foundation (CNCF) that are being established as de facto standards in their areas and offering them as managed services. The implementations may be proprietary or based on the same open-source projects, but for application developers it is important that they can trust these managed services to support the same open interfaces and protocols. Some examples in the database category are PostgreSQL, MySQL, Cassandra and Redis. Within the observability and metrics-monitoring category, Prometheus and OpenTelemetry are starting to appear as such de facto standards supported by major HCPs.

This positive trend opens the possibility for application developers (typically within network software vendors) to build portable applications that can utilize managed services without locking the application into one cloud environment. This equally benefits operations teams (typically within CSPs) by enabling them to build the knowledge of operating the application once and reusing it in another cloud environment when needed.

In addition, even in scenarios where such managed services may be unavailable – such as on-premises deployments – or not a good fit given the use cases or usage patterns, CSPs can continue to rely on network software vendors to provide the same service as they do now.

Provisioning and configuration of managed services

Careful selection of the managed services that an application will use is only one half of the story. The

other half involves harmonizing and unifying the way network software vendor application developers specify the dependency from the application to the managed services, as well as how the CSPs’ operations teams provision, configure and manage these services.

HCP managed services have two sets of interfaces:

- 1. Service interfaces consumed by applications
- 2. Orchestration interfaces used by orchestrator engines, scripts and so on, to provision and configure the service.

While the first set is based on de facto open APIs, the second set is proprietary to each HCP. In some instances, this can even be the case when a single HCP offers two managed services that provide the same service interface (based on the same open API), but the orchestration of the two is different. In these situations, HCP-managed services that offer the same open-service APIs (such as PostgreSQL, Redis or Cassandra) also provide proprietary and different orchestration APIs for provisioning, configuration and management.

For instance, Amazon has two different managed services that provide the PostgreSQL API – Amazon RDS for PostgreSQL and Aurora PostgreSQL – as well as offering guidance on how to decide which one is the best option based on a list of criteria [2]. It would be an advantage if CSPs were free to choose between the two without requiring the network software vendor to change the application package.

As another example, Google Cloud offers the Cloud SQL service, but it is also developing a new PostgreSQL interface for Cloud Spanner. Providing CSPs with the option of a potential future migration from Google Cloud SQL for PostgreSQL to Cloud Spanner [3] without requiring any change in the application package would be very attractive.

By raising the abstraction level and declaring dependency to a backing service with open APIs (such as PostgreSQL, Redis and Cassandra), application developers can avoid dealing with all the cloud-provider-specific orchestration and

	Terraform	Crossplane
State management	The state is stored in state files and updated by Terraform state command. There is no out-of-the-box and uniform way to handle it in different cloud environments. A typical solution is to use an object store such as S3. Versioning and access control are also closely tied to how the state files are stored.	The state and configuration data are stored in Kubernetes’ API server (etcd) as custom objects. A production-grade Kubernetes deployment is needed to properly manage the state for a production environment. Version handling, access control and so on are managed as part of the custom object APIs.
Drift detection and handling	There is no automatic drift detection and reconciliation. This is done manually by invoking the right Terraform commands.	Continuous and automatic state monitoring, drift detection and reconciliation are achieved by utilizing Kubernetes’ operator pattern [9].
Abstraction and customizability	Abstraction is achieved by using Terraform modules, usually combining multiple resources from the same cloud provider. Multi-cloud abstractions can also be created. There is no strong separation between platform and developer teams.	Abstraction is achieved by using composite resources and compositions. Multi-cloud abstraction is built into these concepts. The level of abstraction is fully controlled by the platform team that is seen as a separate persona from the developer team.
Developer experience	The developer must be familiar with HashiCorp Configuration Language.	The developer can utilize existing knowledge of Kubernetes YAML Manifest files, or Helm, or other tools already used for deploying the application in a Kubernetes cluster.
Surrounding ecosystem	Terraform has a large and active community that provides a comprehensive set of modules and integrations for many cloud providers.	Since managed service is represented as a Kubernetes custom object, any tool within the ecosystem that integrates with Kubernetes for application workloads can also be used for managed services. This makes for a very rich surrounding ecosystem.

Figure 4 Comparison of Terraform and Crossplane

UNIFYING THE ORCHESTRATION OF MANAGED SERVICES FROM MULTIPLE CLOUD PROVIDERS IS THE MAIN GOAL

provisioning details, as well as ensuring that the low-level details of orchestrating managed services do not leak into the application layer. The best approach to doing this is to build a common API layer that is fully controlled by CSPs' internal platform teams. This API layer must fully decouple what an application needs from how the underlying cloud providers deliver it.

When it comes to deciding how to implement this API layer, two broad approaches are conceivable. The first is to use infrastructure as code (IaC) tools and frameworks that predate Kubernetes. HCP proprietary examples are Amazon Web Services (AWS) CloudFormation, Azure Resource Manager and Google Cloud Platform (GCP) Cloud Deployment Manager. The most popular HCP-agnostic example is Terraform [4].

The second option is to utilize a more recent crop of IaC tools that are effectively Kubernetes add-ons that can provision and configure managed services external to the Kubernetes cluster and represent them as custom resources to applications running inside the cluster. AWS Controllers for Kubernetes (ACK) [5], Azure Service Operator (ASO) [6] and GCP Config Connector [7] are HCP proprietary examples that all follow this approach.

Crossplane [8], a CNCF project, aims to provide the same capabilities as ACK, ASO and GCP Config Connector, but also add the capability to compose the underlying managed services into higher-level abstractions that fit the needs of a custom-made platform tailored for the requirements of an organization. This makes it a natural candidate for a common API layer that intends to unify deployment and provisioning of managed services from various cloud providers.

Since unifying the orchestration of managed services from multiple cloud providers is the main goal, we see Terraform and Crossplane as two cloud-agnostic options that have the potential to be used in shaping this API layer. *Figure 4* compares the capabilities of the two from various angles.

Observability of the infrastructure and application

Reliance on selected managed services depends on the ability to observe how they are performing and ensure that they are satisfying the Service Level Agreements. HCPs enable this with the help of dashboards that also make it possible to combine observability data from the infrastructure and the application to create a holistic view, which is essential for the application ops team to gain insights in an efficient manner.

In many cases, the collection of observability data is based on HCP proprietary APIs. However, now that HCPs are starting to make use of de facto standard APIs based on CNCF-provided projects, application development teams can benefit from the observability capabilities that the HCP presents without changing how the data is provided. In most cases, however, the HCP will charge based on the amount of data collected – the number of metric samples or log entries, for example.

In a typical telco application where the number of metrics is high and the scraping interval is low, there are two likely outcomes. Either the observability cost will be high or the observability-related data will need to be filtered or preprocessed before it is provided to the HCP observability solution. This could mean that only metrics that relate to important key performance indicators are forwarded to the HCP solution. In cases where a more detailed view is required, the application would still need to provide its own collecting services for metrics and logs to address additional use cases such as machine learning and general troubleshooting. These decisions would be based on the use case and the scale of the deployment.

In a hybrid or multi-cloud-deployment scenario, the problems within the observability areas are

similar to those within the provisioning and configuration of managed services, with siloed cloud vendor tools and a multi-platform monitoring situation. A unified observability system – often referred to as a single pane of glass – is needed to overcome this challenge. This approach is a win-win for both CSPs and network software vendors, as it enables network software vendor developers to rely on well-defined interfaces that expose application metrics, traces and logs, while CSP ops teams benefit from a consistent way of monitoring the applications, regardless of the cloud in which the application is deployed. Removing the need to train ops staff to work with different observability systems also reduces operating expenses.

Several factors influence the decision about where to host the unified observability system. In the case of a hybrid deployment with most of the workloads deployed in private data centers, it may make sense to host it at the private data center as well (with the caveat that extracting data from the HCP may lead to high data egress costs). In cases where the HCP is used for most workloads, the HCP would probably be better suited for the observability solution. The drawback in this scenario is that data from the private data centers needs to be transferred to the HCP, which could create potential issues relating to sensitive data and other legal aspects.

HCPs typically have the capability to ingest observability data from other cloud infrastructures, which allows CSPs to address the multi-cloud deployment scenario. However, moving this type of data across different HCPs typically entails high data egress costs. Initiatives such as the Bandwidth Alliance [10] aim to cut the cost of data transfer between cloud providers and make the multi-cloud deployment more viable.

There are several external SaaS providers of HCP-agnostic unified observability systems, including Datadog, Dynatrace, Sysdig and Elastic Cloud. Some are mostly based on open-source and CNCF projects, while others are more proprietary in nature. The main advantage of these solutions is that they avoid HCP vendor lock-in even for the observability aspect, making them viable in the

hybrid and multi-cloud deployment scenarios. They typically come with small footprint agents to collect different types of data from the infrastructure and the application. By collecting the data as close to the source as possible, some SaaS providers can minimize or even eliminate the data egress cost.

Security and compliance

When utilizing managed services, there is a clearly defined, shared responsibility between the HCP and the application. While the HCP is responsible for protecting the infrastructure so the CSP can offload the burden of operating, managing and controlling the managed service, it remains the responsibility of the application owner to make sure that the managed service complies with applicable laws and regulations. The application owner is also responsible for managing the data, classifying the assets and encryption. This responsibility includes consideration of aspects such as laws and regulations as well as the need for data anonymization related to observability data.

With regard to the integration of HCP-provided security services, the first step is to make use of identity and access management (IAM) capabilities, as this is the way applications gain access to managed services. HCPs have harmonized this by using the Kubernetes concept of a service account that is mapped to the HCP IAM service to provide access to the managed service. From here, deeper integration toward the provided security services can be achieved. The integration would fully utilize and depend on the IAM and key management capabilities that the HCP provides for any authentication and authorization needs, including those between application-specific services. This

SEVERAL FACTORS INFLUENCE THE DECISION ABOUT WHERE TO HOST THE UNIFIED OBSERVABILITY SYSTEM

may, however, come with some challenges with respect to private keys and root of trust, including the determination of which party is liable for keeping them secure.

Looking further into access control, and especially authorization, different managed and application services typically support a variety of protocols. Different protocols may be required depending on the HCP context in which the application is running. The lack of harmonization within this area forces the applications to be flexible about how the authorization should be addressed. Major alignment is not forecast in this area because several different standards already exist, which means that portability would entail an additional cost for application implementation.

Governance and cost management

No matter which underlying cloud environment is used to deploy the applications, governance and various corporate policies need to be enforced in a consistent way. These policies are typically company specific and could impact various aspects such as selection of services based on cost, availability, performance and other considerations. Each HCP typically offers proprietary services and recommendations for monitoring and optimizing costs, but in a multi-cloud deployment, it is important to harmonize these aspects across different clouds and take the CSP's policies into account. CSPs can either build their own cost management layer or utilize commercial cloud cost management offerings such as Apptio Cloudability and Kubecost.

Conclusion

Communication service providers (CSPs) are partnering with hyperscale cloud providers (HCPs) to deploy 5G services where it makes the most business sense and is mutually beneficial. The result will be a hybrid and multi-cloud future, where 5G services need to be deployed and operated across a wide range of public and private cloud environments. Adopting a uniform strategy in developing, deploying and operating these services

CSPs CAN CONTROL TOTAL COST OF OWNERSHIP AND ENFORCE UNIFORM GOVERNANCE AND COMPLIANCE

across such diverse target environments is beneficial to both network software vendors and CSPs. By leveraging open source and using HCP managed services that are compatible with their open APIs, network software vendor application developers can build multi-cloud-native applications that are portable across clouds, while utilizing HCP managed services. This, in turn, results in faster time to market for CSPs. On the operations side, CSPs can control total cost of ownership and enforce uniform governance and compliance by unifying important operational aspects such as deployment, security and monitoring, while partnering with multiple cloud providers.

To achieve success, it is important for CSPs and network software vendors to collaborate and promote the development of open APIs for the platform services that 5G applications need and advocate for the adoption of those open APIs by HCPs. The Cloud Native Computing Foundation has demonstrated that it is one of the best positioned communities for such open collaboration and alignment.

Further reading

- » Ericsson, Cloud native applications, available at: <https://www.ericsson.com/en/cloud-native>
- » Ericsson Technology Review, Cloud-native application design in the telecom domain, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/cloud-native-application-design-in-the-telecom-domain>

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THE AUTHORS



Antonio Alonso

◆ is a technical expert who focuses on cloud-native stateful backing services at Business Area Digital Services. He joined Ericsson in 1995 and has worked on identity and subscription management solutions for different access generations (3G/4G/5G) and domains (circuit switched and packet switched, IMS and identity federation). He joined Ericsson's common development/Application

Development Platform (ADP) organization in 2020, where he serves as the technical leader for the data management, backup & restore and life-cycle-management functional areas. Alonso holds an M.Sc. in telecommunications engineering from Universidad Politécnica de Madrid in Spain.



Henrik Saavedra Persson

◆ is a senior principal developer at Business Area

Digital Services, where he drives the common cloud-native architecture for the business area's network functions and applications in his role as ADP chief architect. He has been a speaker at KubeCon, NSMCon (Network Service Mesh Conference) and the Open Network Summit, presenting Ericsson's approach to cloud-native architecture and the Cloud Native Computing Foundation. He joined Ericsson in 2004 and has worked in R&D with architecture for both applications and platform products. Saavedra Persson holds an M.Sc. in software engineering from Blekinge Institute of Technology in Sweden.



Hossein Kassaei

◆ is a principal developer and lead platform architect, advocating cloud-native-architecture principles, technologies and best practices in Business Area Digital Services. He joined Ericsson in 2010 and has worked in various software and system design and architecture roles in platform and application teams. Kassaei holds an M. Sc. in computer science from Concordia University in Montreal, Canada.

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Creating autonomous networks with intent-based closed loops

To create intelligent networks with the ability to adapt to new situations and changing needs, it will be necessary to decouple requirements from solutions. Intent is the expression of the requirements that an autonomous system needs to meet, which makes it a key concept in the creation of intelligent networks.

JÖRG NIEMÖLLER,
RÓBERT SZABÓ,
ANDRÁS ZAHMSZKY,
DINAND ROELAND

Intent-based operation is a new paradigm for telecommunication systems that is essential to the creation of autonomous networks.

■ Intent is a declarative information object that defines the requirements that an autonomous system and infrastructure are expected to fulfill [1]. An intent is never imperative: the receiving system is not instructed to perform a particular action or process. On the contrary, the system is free to choose a solution strategy autonomously. Intent allows the system to understand global utility and the value of its actions. Consequently, the

autonomous system can evaluate situations and potential action strategies rather than being limited to following instructions that human developers have specified in policies. This means that intelligent decisions that were previously made exclusively by human policy developers can become automated. While intent builds the foundation by providing information about requirements and utility, the implementation of autonomous capabilities remains a challenging task. We believe it can best be achieved by a combination of artificial intelligence techniques such as machine learning and machine reasoning using knowledge-centric architecture and processes

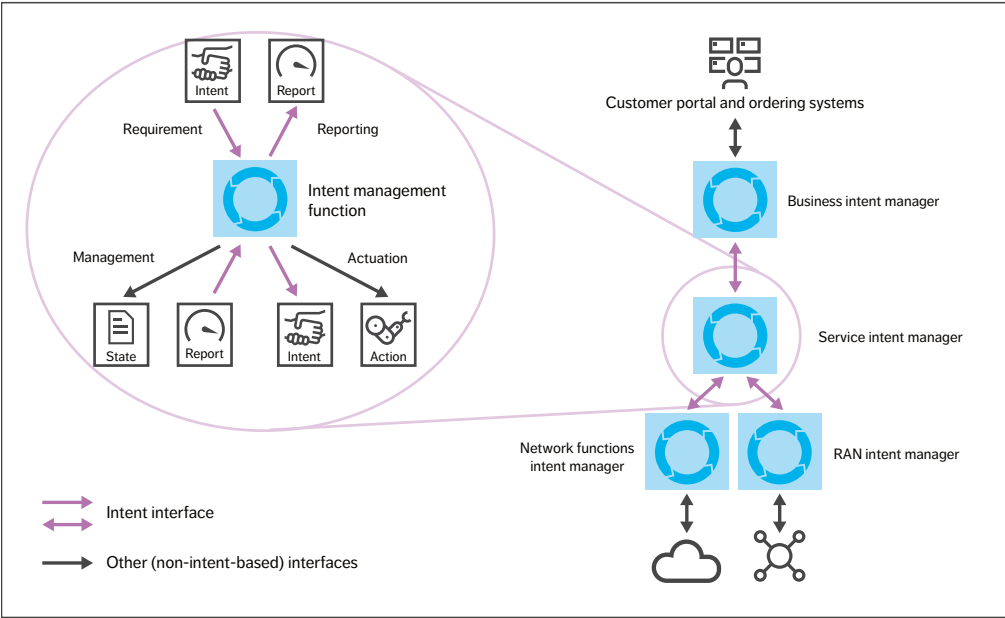


Figure 1 The intent management function and intent-based networking system stack

[2, 3, 4]. The result is a cognitive network based on the vision of 6G as an innovation platform for emerging use cases and applications [5].

Concepts of intent-based autonomous operation

The TM Forum Autonomous Network Project describes a reference architecture for network operation software that is able to act autonomously [6]. It defines a new function called the intent management function (also known as the intent manager) to coordinate intent-based operation [1]. Every autonomous domain contains an intent manager. This is the endpoint of the intent interface for managing the life cycle of the intent [7]. It also coordinates the operation within the autonomous domain to meet the requirements that the intent formulates. This implies that it is able to detect whether the intent has been fulfilled and, if not, find and execute corrective actions.

The left side of *Figure 1* shows the main interactions of the intent management function within a layered operation infrastructure. The intent management function receives all intents directed toward its autonomous domain. It also reports back to the intent origin regarding its success in fulfilling the intent, which closes an intent-based control loop. The intent manager is aware of the system state through measurements, analytics and other information systems such as inventories. Comparing the state to the intent shows if and where the system is not meeting the requirements and indicates whether or not corrective action is needed. As soon as the intent management function finds a preferential action strategy, it executes on it. It can act through conventional management interfaces or, if the targeted autonomous domain is intent-aware, it can act by defining its requirements through an intent. At the highest level, intent-based operation could

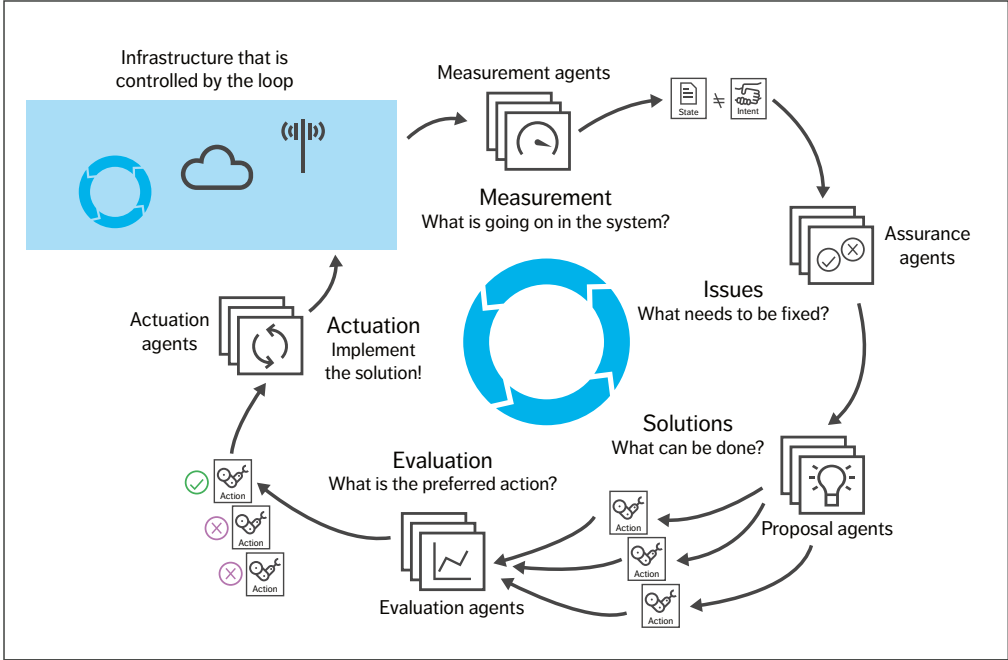


Figure 2 The cognitive intent management loop driven by machine reasoning

start with customer and operator requirements. Autonomous domains coordinated by their intent management function would then make partial decisions resulting in a more specific intent being issued to the next level of operation. This process ultimately arrives at a detailed distributed configuration of the network capable of meeting the high-level requirements that initiated the process.

The right side of Figure 1 provides an example of three levels of intent-based operation. The intent manager at the service level is part of a service management and orchestration (SMO) system. In this example, the solution at the service level has impacts on the network function management as well as on the RAN.

Cognitive intent management loop

The intent control loop between autonomous domains is covered by standardization [1, 7]. The

remaining challenge is how the intent manager interacts with the specific business logic within an autonomous domain, so that they can together translate an intent that expresses business needs into detailed technical configurations.

Figure 2 illustrates how we have implemented an intent management function. We have established a generic cognitive loop that consists of five components: measurement, issues, solutions, evaluation and actuation.

Measurement of the system state is a continuous activity used both to set goals (expressed as intents) and to evaluate fulfillment of intent. The intent manager needs to know which resource instances are used to fulfill the intent as well as their operational state. Knowledge of the current performance of service components at their resource instances makes it possible to determine if the system is in breach of its intent and needs corrective

action. However, as the intent can change, so can the required measurements. The intent manager must adapt its measurements accordingly by employing measurement agents (specialist implementations of measurement tasks) as required.

The issues stage focuses on identifying what needs to be fixed. Issues are defined as requirements (expressed by intents) that are not met by the system. Examples include a required QoE metric that is not reached or a required resource that is not available. Assurance agents implement continuous monitoring of state versus requirements and raise issues when required. This can include a root cause analysis for a more precise description of the issue, prioritization of issues depending on their severity or waiting for ongoing actions showing results.

In the solutions stage, proposal agents react to issues and determine the available corrective action strategies. Any number of differently implemented proposal agents can be used at this stage, including human designed policies and machine learned policies that derive solutions from evidence data. Multiple action strategies may be proposed, each of them representing what the system can do to address the issue.

In the evaluation stage, evaluation agents determine the expected impact of an action on the system state to determine which of the proposed solutions can deliver the most positive effect on the fulfillment of all currently valid intents. Predictive models and digital twins will play a major role in this step, as they make it possible to virtually explore actions and their expected outcomes. The preferred action proposal is the one expected to maximize global utility by best fulfilling all intents. The evaluation stage can also include detection of conflict from actions that fulfill one intent at the expense of degrading another. Evaluation can therefore be a sanity mechanism with the potential to save the network from risky actions and degradations.

A solution with an overall preferential evaluation proceeds to the actuation stage. Intent-based systems can act by using an intent to define requirements on the autonomous subordinate

THE ENTIRE INTENT MANAGEMENT LOOP IS BASED ON AGENTS CREATING AND CONSUMING KNOWLEDGE

system layer. Alternatively, they may act through traditional interfaces by changing configurations of invoking processes. Specialized actuation agents are available to implement a different type of action-taking. For example, in service intent management, a proposal may be expressed by a TOSCA (Topology and Orchestration Specification for Cloud Applications) model and the actuation would therefore be orchestration.

The entire intent management loop is based on agents creating and consuming knowledge. Agents both react to knowledge changes and create knowledge. For example, proposal generation is initiated by the appearance of the type of issue that the proposal agent was implemented to handle. It delivers a proposal for the issue through the creation of a proposal description in the knowledge database.

Intents are introduced in standards [8, 9] as ontology graphs. To enable a complete knowledge-centric implementation of the intent management loop, Ericsson has defined additional ontologies for knowledge about state, issues, proposals and predictions. This allows agent coordination through a generic logical-reasoning-based implementation. This loop is highly self-adaptive to changing intents. As long as respective agents are available, the machine-reasoning-based intent manager will autonomously utilize them by composing a custom, ad hoc workflow.

Within an SMO system, apps would typically be used to implement domain-specific business logic. They constitute agents and as such participate in intent management.

The cognitive intent management loop in Figure 2 is not specific to any particular autonomous domain;

●● IN OUR EXAMPLE, WE
FOCUS ON INTENT-BASED
MANAGEMENT OF A CLOUD-
NATIVE FUNCTION ●●

it is, rather, a template for implementing any intent manager. The creation of an intent manager for a particular autonomous domain is primarily an implementation task of domain-specific agents.

Identifying solutions through resolution

Based on the concept presented in Figure 1 and the work of the TM Forum Autonomous Network Project, we have designed an autonomous network with a multilayered architecture. Operation starts with the intent expressing abstract business requirements – a cost-efficient video service shall be delivered to the silver-level user group, for example. This specifies the subject of a video service, high-level preference, such as cost efficiency and the targeted user group as context.

A business intent manager would receive the intent with business requirements. It would need to translate this into an intent for a service intent management function. This translation, or resolution, would include selecting eligible service blueprints from onboarded service definitions. They would be further contextualized by interpretation and transformation of the abstract requirements into more concrete goals. Policies and apps would typically implement the transformation business logic. In our simple example, the customer group would imply certain locations where the service would need to be available. It would further translate the business requirements of the service into an expected user experience with goals about video quality and availability.

An intent management function associated with service management would receive these requirements about the needed service within an intent from the business intent manager. The main objective of the service intent manager would be the

delivery of a service instance that fulfills this intent. A key aspect of this process would be the distribution of application and network function components over the network and cloud infrastructure. A solution proposal within the intent management loop on the service layer would describe the full topology of these artifacts.

Operator concerns, such as resource utilization, energy consumption and cost, would be considered by the policies that determine the solution. These concerns may also be expressed by an intent and constitute additional operator requirements.

In the solution-finding process, the requirements on the service would be further translated into lower-level requirements per component. For example, a network function would be required in a particular location with goals on latency and bandwidth. This would contribute to delivering the required user experience.

Executing the solution at service level typically involves a service orchestrator that coordinates the execution of distributed actions. Some of these actions may involve sending further intents into autonomous domains that are intent-aware and contain an intent management function. In our example, we focus on intent-based management of a cloud-native function. A cognitive intent management loop such as the one in Figure 2 also coordinates the operation at this intent management level.

The proposal agent used to determine a solution strategy is mainly concerned with identifying all atomic deployment function artifacts and allocating them in the data center. Deployment function candidates are still technology agnostic, but we assume that there is a direct mapping of each atomic function to some deployment artifacts that are understood by the orchestrator or virtualization manager of the administrative domain.

The deployment function candidates may contain other functional dependencies such as the existence of an operations and maintenance function. These dependencies would be recursively resolved until the process reaches a set of atomic functions that satisfies all dependencies and their connectivity

requirements. The process may also involve functional decomposition, when a logical function is substituted with a set of atomic components. For example, a gateway function is decomposed into user plane and control plane components, each adhering to some resiliency and scaling requirements.

Ultimately, the multiple layers of resolutions for a new business intent provide a service instance design that is broken down to locations (such as data centers and transport paths), deployment artifacts (represented by Helm charts, for example) and initial resource allocation with respect to bandwidth, QoS class, virtual central processing units, memory, storage and so on.

Solution proposals in each layer may come from a distinct set of different resolution agents, or a resolution agent may be able to propose multiple solution alternatives. In the next step of the cognitive intent management loop, the most preferential solution may be chosen by an evaluation logic in each layer.

Once the service is instantiated according to the preferred solution, information becomes available that enables service monitoring. This means the intent management loop becomes an assurance loop that takes corrective actions if the intent requirements are not matched by the measured system state.

Conflict detection and resolution

Typically, the solution that a proposal agent proposes aims to improve one metric, which can be problematic in cases with multiple requirements in a single intent or multiple simultaneous intents. As network resources are shared and limited, a proposal that aims to solve an issue related to one metric (when a requirement is not fulfilled) may have a negative side effect on another metric, resulting in unstable network behavior. Overcoming this challenge requires the system to consider the effect of any given action on all the metrics under evaluation, as opposed to just the one that caused the issue. In situations where multiple solutions to a particular issue are proposed, the intent manager

needs to decide which action to approve.

Our implementation includes a conflict detection and a conflict resolution component. Conflict detection examines the outcome of a given action in terms of its effect on all the metrics in the system. If any of the metrics that were met before the action are likely to be violated if the action is taken, those potential negative side effects are considered to be conflicts. In the conflict resolution step (part of the evaluation in Figure 2), the intent manager evaluates whether it should approve or reject the action.

In the prediction step (also part of the evaluation in Figure 2), prediction agents estimate the effect of the given action on all the metrics in the system. In one option, the prediction is achieved by machine reasoning rules inserted into the system by domain experts. The rules are formulated through experience and highlight key correlations between action and measured metrics. For example, changing user plane priority for some user equipment (UE) means that the bandwidth will be shared differently among all the UEs. Consequently, a change in bandwidth means a change in QoE for a certain application. From this, it can be concluded that changing the user plane priority in the network will lead to a change in the QoE. Based on historical measurements, it is also possible to predict how much the QoE is expected to change. Another option for predictions is to utilize machine learning and train models that predict the outcome of the actions.

Next, an evaluation agent examines the proposed actions and decides which one to take. It uses the intent manager penalty to determine whether to execute the action. The intent manager penalty is calculated as the sum of the penalties of all the

●● OUR IMPLEMENTATION
INCLUDES A CONFLICT
DETECTION AND A
CONFLICT RESOLUTION
COMPONENT ●●

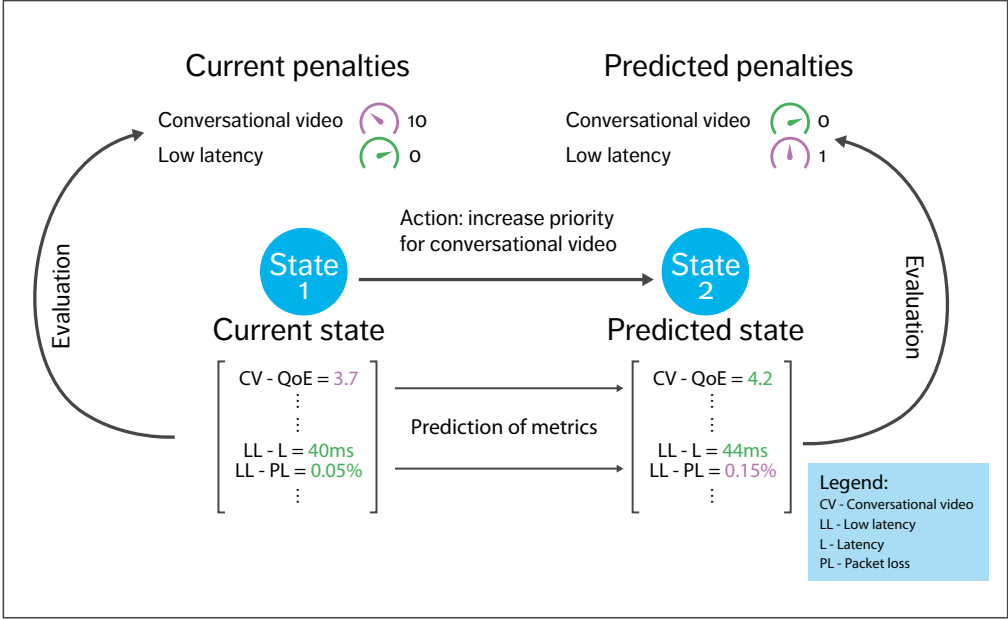


Figure 3 Conflict detection and resolution example

intents handled by the given intent manager. The penalty formula, which was submitted alongside the requirements of an intent, is used to calculate the penalty of the intent. The penalty formula incorporates the cost of not fulfilling the intent.

In its simplest form, it is a constant value, if at least one requirement is not fulfilled, and zero otherwise. In its more sophisticated form, the formula gives a higher value if the gap between the measured value(s) and the target value(s) of the metric(s) is high, and a lower one if the gap is low. As in the prediction step, the impact of each action on all the metrics is determined, so that the evaluation agent can calculate the expected intent manager penalty for each action. An action is approved if the intent manager penalty is not expected to increase after executing the action. In an extreme case, when all the proposed actions are predicted to increase the intent manager penalty, none of the actions will be approved.

The evaluation agent can give feedback to the proposal agent on the outcome of the action. For example, if an action was rejected, the proposal agent can incorporate this information to come to a better proposal in the next round. However, if after multiple attempts, the target of the corresponding metric is still not reached, the intent manager can utilize the escalation process to signal to the submitter of the intent that it cannot be fulfilled.

Figure 3 illustrates an example use case that features conversational video and low-latency services. The first intent expresses that a conversational video service is required and for 80 percent of the users the QoE should be higher than 4.0 (on a scale of 1 to 5). The second intent states that a low-latency service is required, where for 99 percent of the UEs using the service the packet loss should be less than 0.1 percent and the latency below 50 milliseconds. For the first intent, the penalty is 10, while for the second, it is 1. In other words, it is more

important to be able to fulfill the intent for the conversational video than the intent for the low-latency service, in cases of resource shortage situations.

Based on the assumption that only the conversational video service is active at the start, and all requirements are fulfilled, the intent manager penalty in the system will be zero. Then, at a later point, the intent for the low-latency service is added. After the required functions are deployed, and the UEs start to use the service, the target of the metrics for this service on packet loss and latency are met. However, as a consequence of sharing the limited network resources, the QoE for the conversational video users drops and the metric's current value is below the target. At this point, the intent manager penalty is 10 because one requirement of the intent for the conversational video is unmet. This situation is shown on the left side of Figure 3.

The fact that the QoE requirement is no longer being met invokes a proposal agent that is capable of proposing solutions to QoE issues. After considering the alternatives, it proposes to increase the user plane priority for the conversational video service.

Next, in the conflict detection step, a prediction agent runs the rules to see if the proposed action will have any side effects, and it discovers it will no longer be possible to meet the packet loss target for the other service if the proposed action is taken. The predicted state is shown on the right side of Figure 3.

The evaluation agent is invoked to resolve the conflict. It concludes that taking the action would change the intent manager penalty from 10 to 1, and therefore it decides to approve the action.

Conclusion

Our research indicates that a knowledge-centric and machine-reasoning-based implementation of intent management functions in networks offers considerable advantages with respect to dynamic self-adaptation to new situations and new requirements as they arise. This approach has the potential to achieve advanced levels of autonomy, far surpassing what would practically be possible with policy-driven automation. Agents have the ability to

THE PROOF OF CONCEPT CLEARLY DEMONSTRATES THAT THE TECHNOLOGIES NEEDED TO CREATE HIGHLY CAPABLE AUTONOMOUS NETWORKS ARE ALREADY AVAILABLE TODAY

find solutions in complex network topologies and make evaluations based on digital-twin-style prediction models. The proof of concept that we have successfully implemented at Ericsson clearly demonstrates that the technologies needed to create highly capable autonomous networks are already available today.

Further reading

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- » Ericsson Technology Review, Adaptive intent based networking, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/adaptive-intent-based-networking>
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- » TM Forum IG 1253: <https://www.tmforum.org/resources/standard/ig1253-intent-in-autonomous-networks-v1-1-0/>

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THE AUTHORS



Jörg Niemöller

◆ is an analytics and customer experience expert in Solution Area OSS. He joined Ericsson in 1998 and spent several years at Ericsson Research, where he gained experience working with machine-reasoning technologies and developed an understanding of their business relevance. He is currently driving the introduction of these technologies into Ericsson's portfolio of Operations Support Systems/Business Support Systems solutions. Niemöller holds a Ph.D. in computer science from Tilburg University, the

Netherlands, and a diploma degree in electrical engineering from the TU Dortmund University, Germany.

Róbert Szabó

◆ joined Ericsson in 2013 and currently works as a principal researcher at Research Area Cloud Systems and Platform, where he focuses on distributed cloud, zero-touch automation and Network Functions Virtualization. Before joining Ericsson, he worked as an associate professor at Budapest University of Technology and Economics (BME) in Hungary. Szabó



holds both a Ph.D. in electrical engineering and an MBA from BME.



András Zahemszky

◆ is a master researcher at Ericsson Research whose work focuses on core networks and network automation. He is currently leading research prototype efforts in the area of end-to-end network automation. He joined Ericsson in 2007 after completing an M.Sc. in computer science at BME in Hungary.

Dinand Roeland

◆ is a principal researcher at Ericsson Research who joined the company in 2000.

His current research interests involve introducing artificial intelligence technologies into end-to-end network architecture with the goal of achieving an autonomous cognitive network. He has worked in a variety of technical leadership roles including concept development, prototyping, standardization, system management and project management. Roeland holds more than 75 patents and an M.Sc. (cum laude) in computer architectures and intelligent systems from the University of Groningen in the Netherlands.



Holographic communication in 5G networks

Once confined to the realm of science fiction, holographic communication now ranks as one of the most wanted 5G-enabled applications by both consumers and enterprise users, according to recent research.

ALIEL ESSAILI, SARA THORSON, ALVIN JUDE, JÖRG CHRISTIAN EWERT, NATALYA TYUDINA, HÉCTOR CALTENCO, LUKASZ LITWIC, BO BURMAN

Holographic communication refers to real-time capturing, encoding, transporting and rendering of 3D representations, anchored in space, of remote persons shown as stereoscopic images or as 3D video in extended reality (XR) headsets that deliver a visual effect similar to a hologram.

■ After several years of experience with video calls on smartphones and tablets, many users report that they are eagerly awaiting the chance to meet others digitally using immersive communication services such as 3D holographic augmented reality (AR) calls over 5G [1]. Compared with flat screen video, holographic communication can convey the subtleties of non-verbal communication and provide a sense of presence and immediacy that enhances the quality of human interaction. More than 50 percent of smartphone users [2] expect this technology to be available to them within a few years.

Consumers are not the only ones longing for more authentic forms of digital communication. A recent study showed that the main barrier to remote working was the need for social interaction [3]. With the time spent working outside the office expected to increase in the coming decade, many office workers will require more immersive forms of digital interaction.

Starting with AR glasses and holographic calls with spatial audio, office workers around the world expect to benefit from the emergence of haptic technology, which adds tactile sensing of digital objects [3]. More than half of them say they would like a digital workstation with multisensory presence at the office when working remotely. Similarly, in a recent online survey of 7,115 self-defined early adopters aged 15–69 in 14 major cities, 80 percent of the respondents said they expect to have telepresence facilities to socialize with remote colleagues by 2030.

The potential for realizing holographic communication use cases in the years ahead is high. They are already anticipated in numerous consumer and enterprise areas ranging from attending family events as a hologram or meeting a doctor from home to remote presence at the office, expert help at a factory and immersive marketing. As soon as the ecosystem is ready to deliver these new experiences at a good price point and quality of experience (QoE), our research suggests that both private consumers and enterprises will be keen to use it.

The ability to make holographic communication an everyday reality is dependent on three key factors. Firstly, there needs to be a desire that drives the behavioral change (the human factor). Secondly, the appropriate AR devices need to become available. Lastly, mobile networks must have the ability to support the holographic communication pipeline.

The human factor

Users are the primary beneficiary of any type of human communication technology, and holographic communication is no exception. It is therefore essential to understand how they perceive it, along with the benefits from their perspective and their opinions about how to improve the user experience.

The two major user groups for holographic communication (enterprises and consumers) have different priorities. Enterprises will use holographic communication if it satisfies productivity goals better than existing tools. These can be defined as effectiveness, efficiency, and satisfaction in a specified context of use [4].

Consumers, on the other hand, tend to select the communication option that best satisfies their fun goals, characterized by hedonic, emotional and experiential perspectives [5]. Emotions are known to be an important guide to decision making [6]. It is therefore crucial that people's feelings are understood throughout the development process, which is achievable through user studies.

In an internal study on the topic, we discovered that a key emotion people feel about holographic communication is excitement. They are excited about seeing a holographic version of people they

●● A KEY EMOTION PEOPLE FEEL ABOUT HOLOGRAPHIC COMMUNICATION IS EXCITEMENT ●●

know in the same room as themselves, and they are excited about the future of this technology. Such a response is expected due to the novelty of the interaction, but as the novelty (or "halo") effect diminishes, people tend to prioritize factors such as usability, usefulness and familiarity [7].

While some QoE metrics already exist for XR communication [8], several other tools are also available. Design thinking [9] can be used early on to ensure the best solution is built, while qualitative methods such as interviews can be used with early prototypes to provide deeper insights into how people feel. Other human-centered topics such as ethics, consent and accessibility should be considered and regularly monitored. Involving experts in human-computer interaction, ergonomics, psychology and user experience early in the design process will ensure that the final product is in line with the intended users' expectations.

It is essential that the human factor is prioritized in the development of holographic communication. One way to facilitate this would be to start referring to the end-to-end (E2E) pipeline as human to human rather than glass to glass.

Holographic communication

The area of **holographic communication** is divided into three paradigms: (1) avatars, (2) professional-quality digital representations of users and (3) consumer-friendly digital representations of users. "State-of-the-art avatar" technologies use artificial intelligence (AI) to generate a photorealistic representation of a user. Professional-quality digital representations are created through the use of multiple cameras and real-time studio recordings. Consumer-friendly digital representations are generated with the help of an AI-enabled, 3D-capturing setup on consumer grade phones or tablets. The third paradigm is the focus of this article.

	Use case	Typical usage (hours)	Resolution	Battery life	Mobility
Enterprise	Retail shopping	1-4	+++	++	+++
	Logistics (picking)	4-8	+	+++	+
	Augmented education	4-8	+++	+++	+
	Remote support	1-2	+++	++	+++
Consumer	AR gaming	1-2	+++	++	+++
	AR navigation	1-4	+	++	+++
	AR event (stadium)	1-2	+++	++	+++
Enterprise/Consumer	Holographic communication	1-4	+++	+	+++

Figure 1 AR use cases and their requirements on the glasses ranked by importance (+ to +++)

Augmented reality glasses and other devices

Early AR glasses were connected by a cable, followed by the usage of Wi-Fi technology that limited mobility to home and office use in most cases. Over the last five years, however, the AR glasses market has been characterized by strong technology evolution with respect to all of the most important parameters: weight, field of view, resolution, battery life and mobility.

The first generation of lightweight AR glasses, which are connected through USB-C to 4G- and 5G-enabled smartphones, were launched two years ago. The next generation is expected to have built-in cellular connectivity, similar to today’s smart watches. Figure 1 highlights some of the enterprise

and consumer use cases that will benefit most from next-generation AR glasses. There are two different AR device types: those with SLAM (simultaneous localization and mapping) support and those without it. SLAM uses one or more front cameras to create and update a map of the surroundings that allows the device to localize itself and to anchor virtual content (including holograms) on fixed locations in the environment. A true holographic impression is generated by the ability to see the hologram from different angles, which can be achieved with SLAM. The ongoing evolution of both mobile and stationary holographic displays as well as AR-enabled contact lenses will also contribute to

Terms and abbreviations

AI – Artificial Intelligence | AR – Augmented Reality | E2E – End-to-End | FPS – Frames Per Second | IEC – International Electrotechnical Commission | ISO – International Organization for Standardization | LiDAR – Light Detection and Ranging | MPEG – Moving Picture Experts Group | MS – Millisecond | NR – New Radio | QoE – Quality of Experience | RAN – Radio Access Network | SLAM – Simultaneous Localization and Mapping | ToF – Time of Flight | V3C – Visual Volumetric Video-based Coding | V-PCC – Video-based Point Cloud Compression | XR – Extended Reality

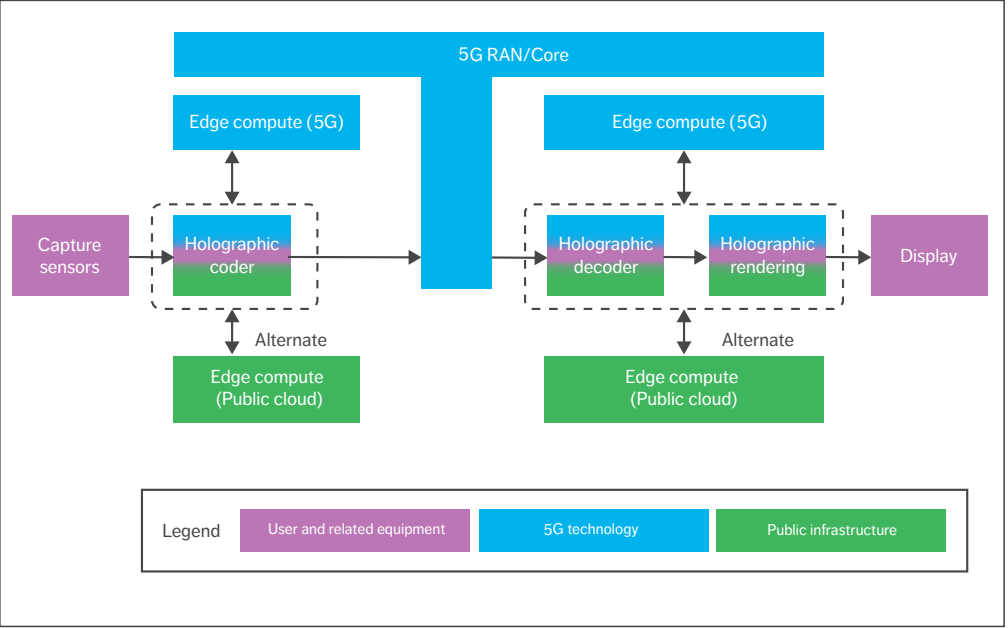


Figure 2 End-to-end pipeline for holographic communication

the uptake of holographic communication. AR-enabled tablets and phones allow a lower barrier to entry in cases where hands-free operation is not required.

Holographic communication pipeline

Our proposed architecture to support holographic communication is depicted in Figure 2. In it, capturing sensors provide a real-time representation of the human face and body. Format conversion and filtering is applied before encoding to reduce the bitrate requirements on the network. The compressed hologram is transmitted to an XR device over a low latency reliable transport network such as 5G. On the XR device, the compressed hologram is decoded and processed before rendering in the consumer environment. The rendering engine considers positioning and semantic information of the device and the rendered scene. The virtual human representation is displayed on the XR device.

The network can act as a compute platform for the encoding, decoding and rendering of holographic data to offload heavy processing from the devices.

Capturing technologies

Holographic capturing is the process of creating a measurable 3D representation of an object, person or environment. This process is divided into four steps:

- 1. Acquisition
- 2. Depth estimation
- 3. Data fusion
- 4. Post-processing.

The acquisition step makes use of visual sensors to capture a volume of interest. Several different visual sensor technologies are used for 3D capturing. Nowadays, the most common mechanism is the usage of time-of-flight (ToF) sensors (such as

●● MESHES CAN BE SIGNIFICANTLY SMALLER THAN POINT CLOUDS, INCREASING THE SPEED OF STORAGE, TRANSMISSION AND RENDERING ●●

LiDAR), which measure distance by the time it takes a light pulse to travel to its destination and back.

In the depth estimation step, sensor streams are used to compute depth. ToF sensors can provide depth information directly, while stereo cameras and multi-camera systems estimate depth by capturing the person from different angles.

In the data fusion step, the depth information or depth map from different perspectives is fused into a single stream of 3D points by matching key points and calculating the optimal geometric transformation from the different views.

The post-processing step reduces the data size of the stream of 3D points by cleaning up redundant points, noise and outliers. The resulting 3D representation can be delivered in various visual media formats such as point clouds or meshes.

A point cloud is a collection of points that represent the captured volume. Each point contains information on its location, as well as the option of the red, green and blue color model and intensity value at a specific frame. A mesh unites those points with triangles, disregarding redundant points and filling any holes. Meshes can be cleaned and reduced further by decreasing the number of vertices. Depending on their resolution, they can be significantly smaller than point clouds, increasing the speed of storage, transmission and rendering of meshes in comparison with point clouds.

Rendering and display

Rendering is the process of generating an image of a scene or a model from a given viewpoint using computations. A scene is a container object describing a volume and its contents. Sources are

objects located in the scene to be rendered. A camera is an instance that renders the viewpoint and consists of a location, focus, orientation and resolution.

Engines render the content based on rendering pipelines. The rendering pipelines take care of culling, rendering and post-processing. There are several types of pipelines with different capabilities and performance characteristics that are suitable for different applications and platforms. A general-purpose render pipeline is optimized to process graphics across a wide range of platforms, while a high-fidelity graphics pipeline is suitable for high-end platforms.

Additional techniques and approaches are available to improve the rendering process in various ways, such as enabling faster rendering with better QoE, smoothing squared edges or increasing the quality of objects in a scene. Furthermore, AI algorithms can recreate an object of a scene or create a photorealistic representation [10] such as an avatar. The photorealistic representation contains a model: a visual representation or mesh model and configurations. Configurations are the bones or access points of a model. These points are animated during the rendering process with preloaded animations.

Real-time data captured from a depth camera takes more computation for rendering compared with an avatar representation. The rendering of all parts of the mesh is needed for every frame for a hologram, compared with rendering a difference for an avatar, such as updating facial expressions like the blinking of an eye on an existing avatar model.

Split rendering is an approach that offloads rendering capabilities to an edge cloud. The key idea is that the six degrees of freedom position and

●● REAL-TIME DATA CAPTURED FROM A DEPTH CAMERA REQUIRES MORE COMPUTATION FOR RENDERING ●●

orientation of the XR device in a scene are tracked, and constant feedback is provided to the edge in real time. The rendering of a 3D scene happens in a cloud. Based on the position of a user, the cloud streams back a 2D video to a scene from the user's angle. For this approach, an end-user device does not need to have high-end characteristics. However, good QoE in this scenario requires low latency communication between the edge and the device.

When it is ready, the rendered stream is transmitted to a device with the ability to provide users with a holographic experience. There are four device types: handheld devices (such as smartphones and tablets), holographic displays, AR glasses and virtual reality glasses. Using these devices, it is possible to place a hologram in a room and move around it.

Media formats and codecs

The delivery of holographic communication will require the processing and transfer of various visual media formats. Those formats aim to represent more realistic and interactive visual representations of humans and/or an environment than established 2D video formats used in traditional videoconferencing, yet the increased informational load may put high pressure on transmission bitrates across the whole communication chain.

For example, point cloud data representing a single human can typically consist of 100,000 to 1 million points and beyond per single time instance (video frame). Streaming such data with 30 frames per second (fps) – the typical videoconferencing streaming rate – would require roughly between 300Mbps to 3Gbps of available bandwidth. Such uncompressed bitrates are not feasible today, and current real-life videoconferencing systems employ media codecs to reduce the bandwidth requirements to single Mbps (1-6Mbps) offering compression ratios from 250:1 up to 1000:1.

Methods to offer similar compression ratios for immersive 3D representations such as point clouds are therefore needed to enable the deployment of holographic communication services. In terms of handling 3D visual formats and their decoding on

●● IT IS POSSIBLE TO PLACE A HOLOGRAM IN A ROOM AND MOVE AROUND IT ●●

XR devices, two scenarios can be considered.

In the first, which is typically applied in the case of split rendering, the processing and decoding of 3D formats is done at the edge, while XR devices decode pre-rendered 2D video with traditional 2D video codecs such as ITU-T (International Telecommunication Union Telecommunication Standardization Sector) and ISO/IEC (International Organization for Standardization/International Electrotechnical Commission) MPEG (Moving Picture Experts Group) High Efficiency Video Coding/H.265 or Versatile Video Coding/H.266 codecs.

In the second scenario, the processing and decoding of 3D formats is done on XR devices requiring the support of additional immersive decoders on the device itself. One approach standardized in ISO/IEC MPEG-I visual volumetric video-based coding (V3C) and video-based point cloud compression (V-PCC) employs 3D to 2D projection algorithms that create intermediate 2D video representations. These representations can be decoded with multiple instances of 2D video codecs such as H.265 or H.266.

An alternative approach would be to employ “native” 3D codecs such as the ISO/IEC MPEG-I geometry-based point cloud compression. In this codec, compression tools operate directly on a 3D point cloud representation. Unlike for the V3C/V-PCC codec, this approach would require adding support for such native point cloud codec(s) in mobile hardware chipsets.

Test results

To determine the feasibility of using existing 5G technology to build an E2E pipeline that enables high-quality holographic communication, we tested the approach presented in Figure 2 in a live 5G New Radio (NR) network.

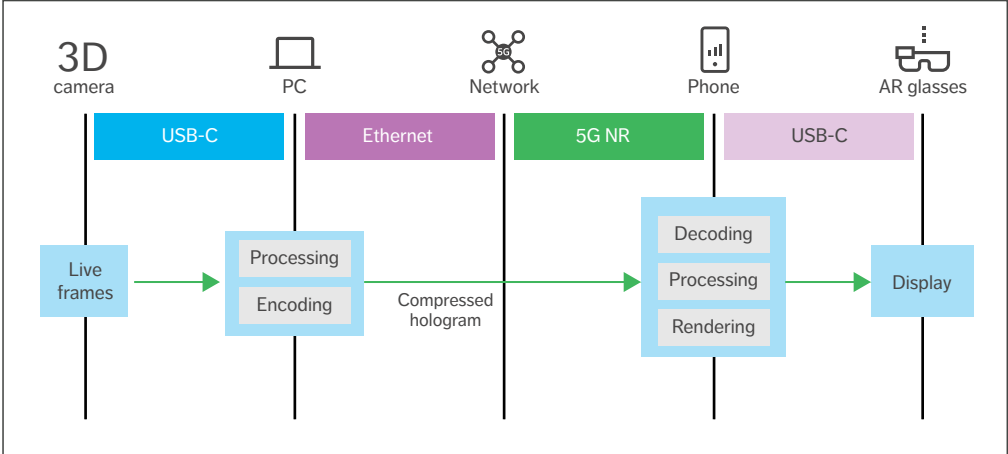


Figure 3 The test setup in the lab

The 3D frames were captured by a single 3D camera connected to a computer, which was connected by ethernet to the 5G network. The captured stream was compressed and sent over the 5G network to a 5G mobile phone with connected AR glasses. The decoding and rendering were executed on the mobile phone and displayed in the AR glasses. The SLAM functionality made it possible to walk around the 3D representation. The measurement setup is shown in [Figure 3](#).

Factors that affect QoE in holographic communications include bandwidth, latency and quality. These three are collectively referred to as the trade-off triangle because only two of them can be optimized at the same time, and it will always be at the cost of the third. In our measurements, we chose to change the quality and bandwidth.

●● FACTORS THAT AFFECT QOE IN HOLOGRAPHIC COMMUNICATIONS INCLUDE BANDWIDTH, LATENCY AND QUALITY ●●

The quality was changed by adjusting the number of points in the point cloud (10,000 points/30fps to 1 million points/30fps). The compression rate was kept constant resulting in bandwidth of between 5Mbps and 100Mbps.

The total latency in the phone is determined by two factors: the decoding and the rendering. As expected, the latency increases when the quality and bandwidth increase. It is clearly visible that the compute in the mobile phone is reaching its limit for the high quality, causing a latency of approximately 170ms (milliseconds). The first experiments show that a move of the decoding compute from the phone to the edge cloud can reduce the latency to 70ms.

[Figure 4](#) shows the bitrate, round-trip time and phone latency results for four quality levels (Q1-Q4) all with 1024x780 resolution and 30fps. The average number of points for each was Q1 (900,000), Q2 (225,000), Q3 (100,000) and Q4 (36,000).

Conclusion

Our research indicates that both consumers and enterprises understand the potential of immersive, augmented reality (AR) applications, and holographic communication in particular, to improve their lives not only in terms of workplace productivity,

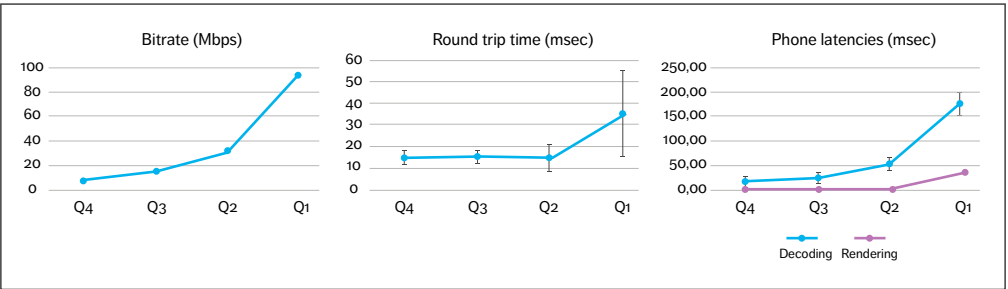


Figure 4 5G NR measurement results

but also when it comes to social interaction and entertainment. The emergence of lightweight AR glasses and powerful 3D compression algorithms have made it possible to start deploying AR use cases using existing 5G technology.

An architecture with edge compute functionality across the full chain is necessary to deliver the 3D captured streams and the rendering of such applications anywhere, while simultaneously meeting device requirements on size, weight and energy consumption. This approach makes it possible to move high performance computing to the network and thereby reduce both the energy consumption of mobile devices and the end-to-end latency.

The introduction of new media formats such as point clouds into extended reality (XR) communication scenarios will greatly enhance the attractiveness, usefulness and efficiency of the information conveyed between communicating parties. Successful communication requires a joint understanding of the communication format, which is typically solved through standardization. In Release 18, the 3GPP has taken on the challenging task of enhancing 5G to offer more efficient support for XR services. Since various aspects of XR are already being addressed in other standardization forums, the intention is to reuse as much as possible.

Further reading

- » **Ericsson Technology Review, XR and 5G: Extended reality at scale with time-critical communication**, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/xr-and-5g-extended-reality-at-scale-with-time-critical-communication>
- » **Ericsson Technology Review, Versatile video coding explained**, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/versatile-video-coding-explained>
- » **Ericsson blog, Welcome to XR immersive experiences**, available at: <https://www.ericsson.com/en/blog/2021/11/welcome-to-xr-immersive-experiences>
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THE AUTHORS



Alvin Jude

◆ joined Ericsson in 2014 where he currently serves as a senior researcher at Ericsson Research. Jude holds an M.S. in computer science from Baylor University in Waco, Texas, USA. He specializes in human-computer interaction.



Natalya Tyudina

◆ joined Ericsson in 2019 and currently serves as a technology developer at Business Area Digital Services. Her work focuses primarily on XR and Infrastructure-as-a-Service. Tyudina holds an M.Sc. in information systems and technologies from Don State Technical University in Rostov-on-Don, Russia.



Lukasz Litwic

◆ joined Ericsson in 2007. In his current role, he is the research leader of the Visual Technology team at Ericsson Research. Litwic received an M.Eng. from Gdańsk University of Technology, Poland, in 2005 and a Ph.D. in electronics from the University of Surrey in Guildford, the UK, in 2015.



Jörg Christian Ewert

◆ joined Ericsson in 1999 as a system manager. In 2005, he joined the core network product management team. At present, he is responsible for 5G communication services in Business Area Digital Services. Ewert studied business administration at the FernUniversität in Hagen, Germany, and holds a Ph.D. in physics from the University of Göttingen, Germany.



Bo Burman

◆ joined Ericsson in 1996. After 14 years at Ericsson Research Media Technologies, he now serves as a senior specialist at Ericsson Digital Services. Burman holds an M.Sc. in computer technology and engineering from the Institute of Technology at Linköping University, Sweden.



Héctor Caltenco

◆ joined Ericsson in 2018 and currently serves as a master researcher in device XR technologies within Ericsson Research. He is also an adjunct senior lecturer at Lund University. Caltenco holds a Ph.D. in biomedical engineering and an M.Sc. in intelligent autonomous systems from Aalborg University, Denmark.



Sara Thorson

◆ joined Ericsson in 2019 and currently serves as head of concept development at Ericsson Consumer and Industry Lab. Her work focuses on emerging use cases in XR and 6G, with a special interest in sustainability. Thorson holds an M.Sc. in international marketing from the School of Business, Economics and Law at the University of Gothenburg, Sweden.

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FUTURE NETWORK TRENDS DRIVING UNIVERSAL METAVERSE MOBILITY

BY: ERIK EKUDDEN, CTO

Expectations about future critical network capabilities and performance have risen significantly among stakeholders across business, societal and governmental spheres in the past couple of years, largely due to the solid performance of the digital infrastructure during the pandemic. At the same time, our industry has continued to invest in evolved 5G and 6G research for a trustworthy and universal open connectivity platform that accelerates digitalization for enhanced social, economic and environmental outcomes, in accordance with the United Nations Sustainable Development Goals.

The Indian industrialist and philanthropist Ratan Tata once said: “If you want to walk fast, walk alone. But if you want to walk far, walk together.” In my opinion, his words are more applicable today than ever before. To enable greater collaboration in the years ahead, we need to build an amplified ecosystem with enhanced development capabilities supported by open interfaces, open source and open innovation. This amplified ecosystem will drive the integration of communication and compute resources into a continuous intelligent network compute fabric, where processing and intelligence can be offered anywhere. System energy efficiency is a fundamental design

optimization parameter of the network compute fabric.

Aside from the technical solutions, the ecosystem will also open up new, innovative business models in areas such as applications, advertisements, devices and payments to drive scale and customer efficiency. This includes optimizing consumer willingness to pay for any specific service, from best effort to guaranteed quality of service (QoS).

Extended reality (XR) and immersive communication will shape the coming decade and become a major advantage in the realization of not only completely virtual metaverses but also in the embedding of virtual content into the physical environment. In last year’s trends article, the first trend focused on the technologies behind digital representation

for the convergence of physical and digital worlds. Since then, I have observed huge investments in metaverse-related technologies such as devices, chipsets, development platforms and applications.

This year, I have selected three network trends as the focus of my annual technology trends article: a unified network platform that enables business-driven enterprise innovation; connectivity for an immersive user experience; and the quest for high-performing networks.

Trends #1 and #2 – and the strong link between them – place extreme performance demands on the coming 6G system to support a wide variety of enterprise use cases that will enable enhanced differentiation. And it is those demands that lead directly toward trend #3.

WHAT IS A METAVERSE?

Metaverse is a common expression for the next generation of the internet that is real time, spatial and interactive. The term was originally coined by Neal Stephenson in his 1992 novel *Snow Crash* and its exact meaning today varies depending on who you ask. For some, a metaverse exists in a purely virtual world that humans experience through virtual reality headsets. For others, it has a strong foundation in the physical world with digital overlays experienced through augmented reality or more interactive mixed reality. In a metaverse, you and/or your avatar are able to interact as humans do, looking into each other’s eyes, perceiving body language and even touching each other.

TREND #1

A UNIFIED NETWORK PLATFORM FOR BUSINESS-DRIVEN ENTERPRISE INNOVATION

A unified network platform will both support and drive the development of a wide variety of new, innovative enterprise use cases. Many of them will involve the delivery of easy-to-use immersive experiences by and between humans as well as autonomous communication and operation between intelligent machines or between humans and machines. As such, the unified network platform will act as an open innovation platform connecting different ecosystems, facilitating collaboration and business growth.

One of the keys to enhanced application innovation is the evolution of network user/developer interfaces through exposed, easy-to-use functions. This includes, for instance, functions that enable productivity, communication, collaboration, gaming and enhanced video. The unified network platform will offer aggregation capabilities that enable interoperability across the actors in the ecosystem to secure unified and global reach. These exposed network capabilities will optimize both application performance and user experience. The developer will have the freedom to select the level of differentiation depending on the application/consumer needs such as quality of experience (QoE), performance, security and privacy.

Architecture

One fundament of the unified network platform is the possibility to aggregate and enhance service exposure from different networks. Examples of exposed functions through open application programming interfaces (APIs) include

QoS prediction, spatial information, user identity, service authorization, instantaneous radio coverage and user-density maps. To manage the different service provider network interfaces, the unified network platform will secure aggregation and routing of the exposed network functions. These network functions will be enhanced by integration of other value-adding functions originating from the global ecosystem. This can involve mediation and interoperability functions for spatial rights management, for example.

The network features and services exposed by the unified network platform will enable the developer ecosystem to build and monetize new applications. Exposed services will range from basic aggregated 5G network features, unified communication services, to more advanced services such as guaranteed QoS, real-time gaming, XR conversations and dedicated industry services. Advanced and highly differentiated services that leverage network slicing will be delivered according to service-level agreements (SLAs). The unified network platform will provide a unified software development environment that will lead to a surge in service and application innovation.

The unified network platform will benefit from industry alignment that ensures cost efficiency, faster innovation and industry adoption. In particular, it is important for the industry to align on exposed network capabilities through open APIs for ease of use in the innovative enterprise business development process.





TREND #2

CONNECTIVITY FOR AN IMMERSIVE USER EXPERIENCE

Many network capabilities will need to grow exponentially during the decade ahead in order to unleash the full potential of technologies such as XR, artificial intelligence (AI), the Internet of Things (IoT) and the Internet of Senses. After successfully handling the exponential growth of each previous technology generation, our industry is now investing in evolved 5G and 6G research to meet future requirements.

Industry leadership will require QoE differentiation from the best-effort services that have traditionally dominated the IT industry. Guaranteed QoE requires solutions that span the end-to-end (E2E) ecosystem of device, network, distributed and central compute, and application actors. This calls for collaboration among the different actors in the ecosystem to establish open standards that enable global scale, innovation, interoperability and performance.

Opening the door for extended reality

Starting from more basic functions, XR applications will develop as devices and network capabilities advance. Important application clusters for this evolution involve gaming, entertainment, social communication, retail, shopping and virtual work, for example.

Existing XR applications primarily focus on a single user who is physically present in a predefined static environment with immersive content that is semi-static in the sense that it only partially adapts to the environment, such as attaching to the floor or another flat surface. This will evolve to dynamic environments that contain moving objects and people, which means that applications need to start adapting to such dynamics.

As XR continues to mature, it will eventually be possible for multiple users to be physically present in dynamic environments with content that dynamically adapts to the surroundings. Real-time occlusion of the rendered content will enable a fully spatialized digital experience.

To render the immersive content, the

physical environment needs to be replicated in a digital format known as a spatial map. Spatial maps are built on static physical environmental data, such as real estate and roads, overlaid by real-time physical environmental data such as moving cars and pedestrians.

To master the rendering, the spatial map information also needs to include the location and orientation of the application user, including their head movement and foveal area – that is, the area covered by the part of the human eye that is responsible for high-acuity vision.

Network evolution

XR applications will demand new system design optimization across the E2E system of device, connectivity, edge and cloud. For instance, spatial-map compute and rendering distribution will have a strong influence on device energy consumption, weight and size. Spatial mapping and rendering processing will need to be offloaded in order to design iconic devices with eyeglass-style, slim form factor and long battery life. Our research at Ericsson indicates that processing offload of XR applications to the edge reduces device energy consumption by threefold to sevenfold depending on the level of device processing offload.

The move from traditional 2D media to advanced immersive media services increases the informational load, due to the multiplicity of media streams and the increased media quality requirements. It puts high pressure on processing and transmission bitrates across the whole communication chain asymmetrically depending on how the XR use case is implemented – that is, it can impact the uplink, the downlink or a combination of both. For instance, device spatial-mapping compute offload (to edge/cloud) will result in a more symmetric traffic load in the downlink and uplink compared with mobile broadband (MBB) traffic, which is mainly heavy downlink traffic.

To guarantee QoE for XR applications, stringent bounded latency requirements are needed when device compute is offloaded to the edge and the cloud. To reduce the bounded latency requirements smart on-device processing techniques will be implemented, such as asynchronous time warp that transforms network-rendered content to compensate for pose changes between time of rendering and display.

To optimize QoE for all network users, the traffic for XR applications can be separated from other MBB traffic with the help of intent-based network slicing. Further, to ensure that latency requirements are met, time-critical communication features such as radio-access network (RAN) assisted rate adaptation (using low-latency, low-loss, scalable throughput technology) and latency-optimized scheduling will be introduced.

There is a strong relationship between wide-area cellular network coverage, capacity and latency demands. The key parameters for improving wide-area cellular network coverage are allocation spectrum efficiency and inter-site distance. For 2030, Ericsson Mobility Report forecasts a traffic increase that is higher than the expected spectrum gains. As this will not be sufficient to support the forecasted traffic increase, network densification will grow in importance to ensure capacity and increased uplink coverage for limitless connectivity.

The growing differentiation of XR services and the variety of new device types require more intelligent interaction with the network. In a cognitive network, the orchestration of these interactions involves tasks such as device onboarding, connectivity management and QoS policy selection. The network must have the ability to distribute actions among devices, the RAN, core, edge and application to dynamically secure the QoE with minimal E2E resource utilization. A first step in this direction is the Dynamic End-user Boost developed by Ericsson, a smartphone app that enables the user to dynamically optimize QoE.



TREND #3 THE QUEST FOR HIGH-PERFORMING NETWORKS

As the digitalization process continues to advance, many human experiences are going to become much more immersive. For example, the work-from-anywhere experience will connect colleagues from separate locations as if they are physically present in the same environment. The digitalization of industries will leverage a digital representation of the complete physical operations – known as a digital twin – to improve operational technology applications. The evolution toward fully autonomous vehicles is another excellent example of the ongoing transformation that is happening all around us. Network deployments that deliver dynamic,

adaptable and distributed connectivity services are essential to support all of these advancements.

Moving beyond best-effort connectivity

A key element of future connectivity services is global and pervasive coverage, both indoors to outdoors, that can meet the requirements of society-critical MBB, as well as mission-critical and business-critical applications for enterprises and industries with extreme radio access performance at selected locations. This connectivity also includes seamless service delivery between, for example, private and public networks or terrestrial and non-terrestrial networks.

The connectivity requirements will vary in terms of performance, with some use cases requiring close to zero consecutive packet losses and connectivity that is always on. Such time-critical and bounded latency applications require several times more radio resources than typical MBB served by public networks today. The combination of a high-performing network with reliable and deterministic processing in the edge (or the cloud) will make it possible to offer E2E guarantees to critical applications.

Such high-performing networks also imply network solutions with downtime that is near to zero. It is critical to consider all network nodes and segments to identify and address

weak points to improve the characteristics for E2E services. E2E availability and resilience management will be provided through an application-aware network that dynamically orchestrates the provisioning according to the agreed QoS.

Trustworthy networks

By the end of this decade, billions of IoT assets, products and machines will be connected all around the world. Ensuring efficient, secure and low-cost deployment and operation throughout their 10-20 year lifetimes requires a fully automated process that can handle devices in bulk. The use of cellular credentials such as eSIM and iSIM as a single root of trust

will be key to ensure global reach.

As resources and applications become more distributed, requirements on security are increasing. The need for E2E solutions to meet customer and regulatory security demands is growing quickly. To address this, all identities must be authenticated and authorized in the distributed E2E link of cloud-network-device-application-user. The best way to facilitate this requirement is by implementing a zero-trust architecture and control based on the principle of “never trust, always verify.” Zero-trust solutions heavily rely on intent-based automation built on AI techniques such as machine learning and machine reasoning.

PROOF OF CONCEPT FOR 5G-ENABLED INDUSTRIAL ROBOTICS

Within the Ericsson 5G-SMART project, leading global technology company ABB and Ericsson have demonstrated how large parts of the control of industrial robots can be offloaded to a digital representation of the robot and its surroundings in an edge compute environment that is connected to the (mobile) robot via a high-performing 5G network. Key robotic tasks such as action planning, localization, mapping, motion planning and navigation are executed in the edge. A machine vision system on the shop floor can be connected to the control of a mobile robot in the edge cloud to enable precise robotic actions with sub-centimeter precision to support actions such as docking a mobile robot at a target station and performing pick-and-place operations.

Conclusion

Digitalization offers unlimited opportunities to individuals, enterprises and society by enabling powerful new solutions that provide mission-critical services, immersive communication and the omnipresent Internet of Things. The successful rollout of these use cases – each with its own unique benefits – will have a direct correlation to the performance of the digital infrastructure that they use. Performance is a major competitive differentiator and, as such, represents a first-mover advantage for communication service providers.

Limitless connectivity, trustworthy systems and cognitive networks are the main differentiation pillars of evolved 5G and 6G. Going beyond connectivity, evolved 5G and 6G will become the trusted platform for intelligence, compute and spatial data, encouraging open innovation and serving as the information backbone of society, industry and enterprises.

A unified network platform is the most efficient system to deliver the performance, quality and scale required to enable universal metaverse mobility, as well as being the most affordable communication solution for humans and machines in any reality. Another major advantage is that the digital infrastructure is already an integral part of the digital economy today, providing a broad range of services and business logic to the ecosystem.

At Ericsson, it is our firm belief that the best way to build a trustworthy and scalable digital infrastructure is through business-driven investments and long-term collaborations between ecosystem actors. With this in mind, we are committed to substantially investing in research and development in evolved 5G and 6G, as well as engaging in deep and long-term collaborations with customers, vendors, enterprises, industries and society at large. Together we will make the unimaginable possible, and create a world where limitless connectivity improves lives, redefines business and pioneers a sustainable future.

ABOUT THE AUTHOR

As Group CTO, Erik Ekudden is responsible for Group Strategy and Technology. His extensive experience of working with technology leadership globally influences the company's strategic decisions and its investments in 5G, 6G, edge computing, artificial intelligence, augmented/virtual reality and the Internet of Things. Ekudden's leadership builds on his decades-long career in technology strategies and industry activities. He joined Ericsson in 1993 and has held various management positions in the company, including Head of Technology Strategy, Chief Technology Officer Americas in Santa Clara (US), and Head of Standardization and Industry. He is also a member of the Royal Swedish Academy of Engineering Sciences and the publisher of Ericsson Technology Review.



Further reading

- » **Ericsson Technology Review, Five network trends towards the 6G era**, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/technology-trends-2021>
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Energy-optimized network modernization

Minimizing energy consumption and maximizing the use of sustainable energy sources at ICT sites requires a transition toward a smart energy setup and a holistic approach to energy management that includes close collaboration with the energy utility sector.

ANETTE HÖGLUND,
JOHAN PETTERSSON,
HELENE HALLBERG,
LARS HUMLA, ERIK
SANDERS

The evolution toward a smart energy setup at Information and Communications Technology (ICT) sites is a key enabler for the creation of resilient, robust and cost-efficient radio access networks (RANs). At the same time, this approach also opens up new income streams for ICT site owners to support the electricity power system (power grid) with ancillary services.

■ While energy efficiency has always been an important consideration in network development, 5G includes an exceptionally powerful set of tools that communication service providers (CSPs) can use to minimize network energy consumption. As impressive as the 5G energy narrative is, however, it is only one part of the solution that the ICT industry needs to overcome in its energy challenge. Fully addressing the challenge will require broad action

across the entire ICT network that includes the evolution of the energy setup to ensure alignment with the needs of the smart grid.

There are multiple interdependencies between ICT networks and the power grid. Alignment and collaboration between ICT site owners and power companies will open up possibilities for new services and income for ICT site owners, while simultaneously contributing to a more resilient and robust power grid.

Growing awareness of climate change [1] is driving a mindset shift in the ICT industry toward a more holistic approach to benchmarking the energy capability needs in a network. With this comes a desire to modernize telecom equipment to increase energy efficiency. Beyond using 100 percent renewable energy, the shift also includes new energy perspectives that focus on optimizing the energy setup at sites. The aim is to ensure high network availability for service assurance and enable intent-based energy management with the ultimate goal of achieving net-zero energy cost.

As the ICT network matures, ICT site owners have an ongoing opportunity to reflect on potential improvements from an energy setup perspective. While a wider range of options are available when building new ICT sites, there are many ways to

reduce energy costs at legacy sites as well.

Figure 1 illustrates Ericsson’s comprehensive energy perspective on network modernization where the focus evolves during the journey toward net-zero energy cost. The ICT site owner’s focus will shift from pure energy consumption reductions in the first phase to enabling smart energy in the second, and ultimately gaining new income in the third.

As highlighted in Figure 1, there are five evolution steps toward net-zero energy cost and beyond:

- 1. Explore possibilities to reduce energy consumption in each network
- 2. Build and optimize for a low network energy consumption
- 3. Ensure autonomous, resilient and holistic energy usage in operations
- 4. Build power grid and ICT interdependency to ensure high availability
- 5. Open a new source of income by provisioning ancillary services for the power grid, such as an open energy bid and clearing support.

These steps provide opportunities to reduce energy costs as well as opening up the potential to turn energy generation into a new source of income.

Key definitions

ICT site – a distributed edge compute site with radio communication capabilities. Future sites that are located far out in the telecom network will include the kind of compute capacity that is currently only available in large, centralized data centers.	many countries for ISPs to own ICT sites today. The meaning of site ownership varies from case to case, but it often includes responsibility for supporting equipment such as the shelter, energy setup and climate control, as well as application equipment such as radio and data servers.	is responsible for operating, ensuring the maintenance of and, if necessary, developing the transmission system in a given area.	Energy system setup – a technical solution at an ICT site that provides the applicable type of energy and the interface to the power grid, which can provide energy backup and relies on autonomous central or distributed energy management.
ICT site owner – ICT sites have traditionally been owned by CSPs, but it is increasingly common in	Transmission system operator – a natural or legal person who	Distribution system operator – a natural or legal person who is responsible for operating, ensuring the maintenance of and, if necessary, developing the distribution system in a given area.	Net-zero energy cost – the reduction of an ICT site owner’s energy bill (the opex cost) to zero.

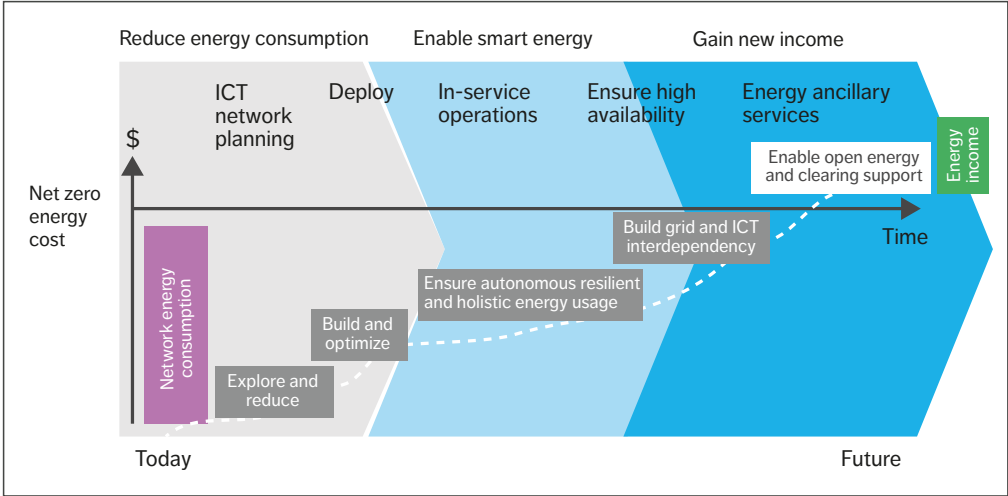


Figure 1 Ericsson's comprehensive energy perspective on network modernization

Approaches to reducing energy consumption

There are four primary aspects that must be considered when reducing network energy consumption:

- 1. Hardware modernization and new software features
- 2. Energy-saving software features and artificial intelligence (AI) enabled automation
- 3. Network energy optimization services
- 4. The support of digital twins.

Hardware modernization and new software features

The biggest reductions in network energy consumption to date have resulted from the

introduction of new hardware and the removal of old systems. Major hardware modernization projects are usually triggered by new releases of spectrum and/or new releases of network technology (such as 5G). Looking ahead, one of the best ways that CSPs and infrastructure service providers (ISPs, also known as tower companies) can speed up their progress toward net-zero energy cost is to consider energy consumption when performing traffic simulations at the network planning stage.

Energy-saving software features and AI-enabled automation

The continuously expanding scope and growing complexity of 5G applications is putting greater demands on networks to deliver high availability, ultra-reliability, low latency and high security. More

automation is essential to cope with these demands. To be successful, this automation must be built on a solid understanding of site capability, capacity, performance, user demand, weather conditions and energy availability at any point in time.

The energy-saving software features in 5G enable CSPs and ISPs to minimize energy consumption during low traffic periods, reduce energy waste during peak traffic hours and turn equipment on and off at exactly the right time [2]. Software functionality with AI agents at the site will improve the CSP's/ISP's understanding of the energy perspective, identify the root causes of inefficiencies and take autonomous action to turn assets on and off when needed.

More advanced software functions will make it possible to reduce manual work, find faults early and maximize network availability. The use of sensors at each site to measure temperature, humidity and electrical characteristics will provide valuable information about the local conditions that make it possible to optimize energy utilization and maximize the life cycle of site equipment.

Intelligent agents capable of handling complex processes are needed to optimize trade-offs between long-term benefits of the agent's behavior and short-term benefits from the immediate steps to be taken; for example, how to optimize a network in multiple steps and how to operate a network. These processes must be learned autonomously, without the intervention of a human domain expert.

Network energy optimization services

Network energy optimization services help CSPs/ISPs explore the full potential of reducing energy consumption in each technology network (4G and 5G, for example). CSPs/ISPs gain the confidence to activate RAN energy-efficiency features by learning how optimization cycles, radio KPI monitoring and the tracking of crowd-sourced consumer experience data ensure that the network's Net Promoter Score remains high. A network energy optimization service can also analyze the system-embedded energy metering and network-wide hardware installed base to evaluate the need for hardware modernization.

DIGITAL TWINS ARE AN IDEAL SOLUTION FOR ICT SITE OWNERS THAT WANT TO AVOID THE EFFECTS OF ERRATIC INITIAL EXPLORATIONS ON LIVE MOBILE NETWORKS

The support of digital twins

Digital twins are an ideal solution for ICT site owners that want to avoid the effects of erratic initial explorations on live mobile networks. A good example of where they can be useful is in the case of network parameters that are configured on a per-cell level but may have a strong impact on the performance of surrounding cells. A change in any of these parameters also affects users served by the surrounding cells.

Finding the optimal configuration for these types of parameters is a complex exercise. Digital twins make it possible to test on an external entity that mimics the behavior of the live network. Once the agent has acquired all the necessary knowledge from the digital twin, the achieved policy can be safely applied to the live network.

Looking ahead, AI will be fundamental in the digital transformation of many sectors as they optimize or introduce new services to use energy more efficiently, to plan and predict maintenance.

Optimizing the energy setup at ICT sites

Energy experts predict more frequent disturbances in the power grid in the years ahead due to extreme conditions such as wildfires, heavy rain and flooding. Increasing reliance on renewable energy sources and the growing popularity of electric vehicles is also expected to have a volatile impact on the power grid. With this in mind, CSPs/ISPs that want to ensure high network reliability without increasing the energy cost should review the capabilities of the

Terms and abbreviations

AI – Artificial Intelligence | CSP – Communication Service Provider | DC – Direct Current | ICT – Information and Communications Technology | ISP – Infrastructure Service Provider | Li-ion – Lithium-ion | PV – Photovoltaic | RAN – Radio Access Network | VDC – Volt Direct Current

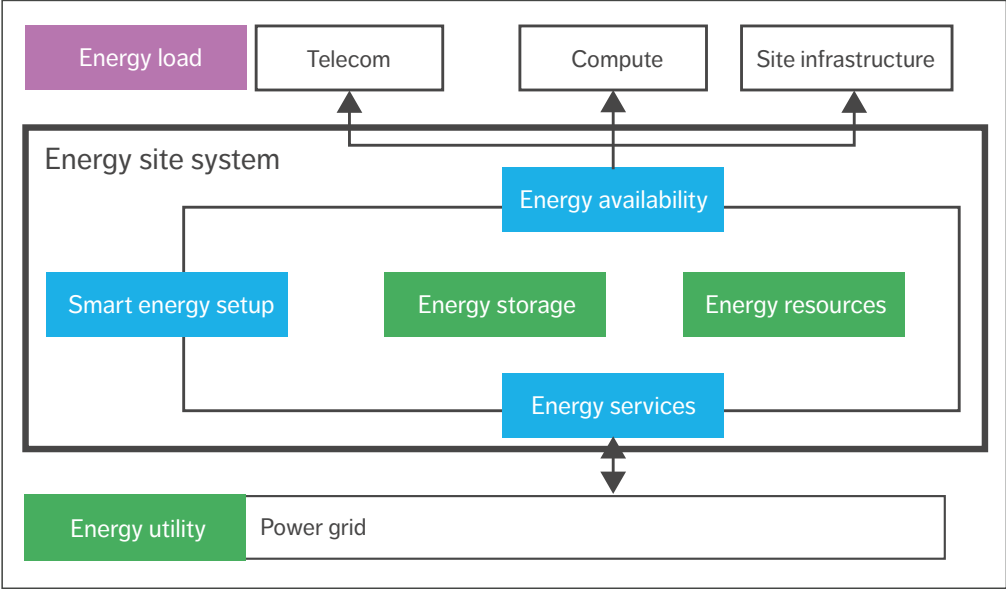


Figure 2 Energy system setup at an ICT site

energy setup at their ICT sites and look for opportunities to improve their resilience.

There are three core actions that ICT site owners can take to ensure the provision of a robust and reliable ICT network, while simultaneously minimizing their impact on the climate and the environment:

1. Introduce local renewable energy generation (solar and/or wind) and add services such as weather forecasting where AI predicts the control of energy storage to optimize the usage of renewables.
2. Improve the overall efficiency of the power system by reducing the number of conversion steps, improving the efficiency/unit and/or increasing system voltage.
3. Establish a smart energy setup capability, where the energy availability at the site is based on the power grid, storage and resource status, and optimize the utilization of site assets.

Figure 2 illustrates the energy system setup that we recommend for ICT sites. The creation of a robust autonomous energy solution makes it possible for ICT site owners to ensure energy availability even at the sites that are furthest out on the edge.

Traditionally the telecom network has only had radio and core equipment at network sites and has relied on data centers for compute capacity. As more advanced use cases are introduced in 5G and beyond, the need for compute at network sites on the far edge will arise, and network sites will turn into ICT sites. The implication of this is that the energy site system needs to evolve in parallel to ensure the allocation of the available energy is done in an intelligent way. A smart energy setup will be key to achieving resilient and sustainable energy usage.

Energy storage and resource technology

The power grid is the most common source in the energy setup at today's ICT sites, with complementary sources utilized when and where

required. At off-grid sites and for sites with seasonal variations, diesel-powered combustion engines are still the main complementary source, although ethanol and liquefied petroleum gas are replacing diesel in some countries. In locations with available space, photovoltaic (PV) silicon panels are sometimes used. While lead acid batteries continue to be the most common choice for backup, batteries based on lithium-ion (Li-ion) technology are becoming increasingly popular.

To make the site energy setup more sustainable, multiple evolutions paths are possible, but some of them may have limitations. To reduce the usage of diesel, fuel cells can be used with fuels in gas, liquid or solid form. Combustion engines will also use fuel variants that have lower greenhouse gas emissions.

The need for Li-ion batteries in the coming years will exceed the production capacity and that will open up for new battery chemistries. PV panels will have improved efficiency by tandem cells or other chemistries. Even if the efficiency of low-cost, thin-film PV technologies is lower than traditional technologies, they could gain market share and be suitable for areas with fewer site restrictions.

Future evolution steps in excessive solar locations will hopefully generate hydrogen that can be stored and used as long duration backups. Combustion engines that can run on hydrogen are also expected to emerge. A future wind turbine design that can harvest low wind speeds while still being robust enough for high winds would make wind turbines a potential contributor to the energy mix at ICT sites.

The energy setup for ICT equipment includes power supply units with remote access and has intermittent peak shaving capability. The energy evolution will provide the hybrid control and intelligence to impact the peak load. The system operational voltage may move from 54VDC (volt direct current) to 57VDC, including DC/DC (direct current) backup. The next step is to evolve from 48VDC to nominal 400VDC. The higher system voltage introduces a reduced size of the cable area and decreases the deployment cost of DC distribution at ICT sites.

THE ICT NETWORK HAS THE POTENTIAL TO PLAY A SIGNIFICANT ROLE IN REDUCING ENERGY CONGESTION AND ENSURING STABLE FREQUENCY

Collaborating for greater resilience and robustness

The transformation of power generation from a centralized model to a decentralized one, characterized by the generation and storage of vast amounts of renewable energy sources, creates new challenges that require reforms to the electricity market. Looking ahead, it will be the responsibility of the transmission system operator [3] not only to ensure resilience and robustness in the power grid, but also to enable interactions in a wider electricity sector/community.

Government strategies will have a big impact on how energy markets can develop flexibility in energy services at international, national and regional levels. Given the opportunity, the ICT network has the potential to play a significant role in reducing energy congestion and ensuring stable frequency.

Collaboration will be required to ensure an optimized and interoperable path that enables adaptive energy consumption by the infrastructure at the ICT sites to contribute to balancing energy services at the national level.

ICT and power grid interdependency

Many of the requirements on the operational performance of ICT networks are similar to those of energy utilities, particularly the new decentralized power grid from the transmission and distributed system operators. Both need to provide capabilities that enable the generation and storage of renewable energy, boost operational efficiency and increase resiliency and reliability.

In terms of renewable energy and storage

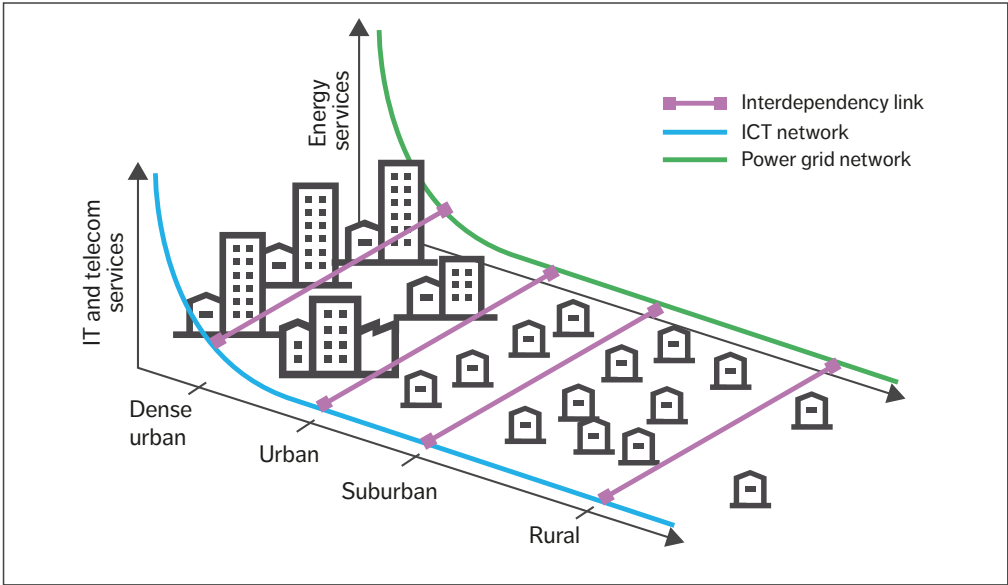


Figure 3 The interdependency between ICT and the power grid

resources, there is a need to adapt and utilize what is available at the edge or local area with small-scale production intermittent availability and reduce/increase energy consumption from time to time. Operational efficiency requires capabilities that optimize assets and utilization, as well as providing high network and service availability. Automation to mitigate service disruptions and power outages and redirect traffic/power flow in case of disasters is key to ensuring resiliency and reliability.

Figure 3 illustrates the interdependency between

SIGNIFICANT MUTUAL GAINS CAN BE ACHIEVED BY ENABLING COMMUNICATION AND SERVICE COLLABORATION BETWEEN THE ICT NETWORK AND THE POWER GRID

the ICT network and the power grid. Significant mutual gains can be achieved by enabling communication and service collaboration between the two. Reliability, availability and resilience are essential to ensure high network availability and service assurance, as well as increasing robustness and improving maintainability, which reduce total cost of ownership in the long run.

The ICT network and the power grid can mutually benefit in their respective service sectors by sharing relevant performance and status information at multiple intersection points, particularly when the situation at sites is challenged by extreme situations like wildfires. This approach enables both networks to maintain service in the local geographical area, ensuring their ability to support local residents and businesses when they are needed most.

Creating ancillary services for the power grid
ICT site owners can start now to prepare for the possibility to support the power grid's needs of different types of ancillary services, thereby opening

up a new source of income [4]. This can be done in five steps:

1. Establish an energy profile for the site and optimize site power system capacity.
2. Introduce a segmentation usage capability of energy storage for different purposes such as ICT network backup and ancillary services. Consider adding storage to increase the income potential from the energy ancillary service.
3. Optimize the ICT site's own energy consumption and local energy production while balancing it with energy storage at the site.
4. Manage the energy consumption of multiple grid-connected ICT sites as one entity to maximize the impact on the power grid.
5. Utilize energy arbitrage by producing energy at the ICT sites and sharing it with the power grid when needed.

An efficient deployment includes AI-based automation to best optimize business intents and support the smart energy capabilities for both the ICT network and the power grid [5].

Supporting the digital transformation of the energy sector

Energy utilities currently use a variety of telecommunication technologies including fixed, fiber, microwave and mobile. In all cases, the communication has to be secure and highly available, as energy services must be reliable and in balance, according to the redundancy policy in operation.

Looking ahead, it is clear that the digital transformation of the energy utility sector would benefit from the introduction of LTE/5G technology [6]. These technologies make it possible for ICT sites to provide energy utilities with insights for the geographically distributed low-voltage power grid with characteristics of outages and performance degradation. Further, with the help of AI agents, LTE/5G networks can help predict performance degradations/deviations and their potential impact.

5G networks are particularly well suited to smart-grid management [7]. The 5G network can be either commercial or private depending on preferences on the energy utility. With a 5G network in place, an energy utility can ensure that its critical functions are implemented in a resilient, reliable and secure way. A 5G network is also the most efficient way for an energy utility to establish a cost-effective strategy for connecting grid assets over a wide geographical area. All new devices in the power grid use wireless technology for communication. Long-term, stable technology evolution is guaranteed, in alignment with the long life-cycle considerations of the telecom segment.

Conclusion

Whenever the need or desire to modernize ICT sites arises in a local or regional area, new energy-related opportunities arise with it. A holistic approach is key to maximize the potential for energy consumption reductions in existing and future ICT networks across the whole service life cycle. To be successful, it is essential that ICT site owners explore and identify candidates to reduce energy consumption as early as possible – ideally in the network planning stage. During the deployment stage, it is crucial to build and optimize for low energy consumption. With respect to in-service operations, artificial-intelligence-based automation, together with active network energy optimizations, make it possible to minimize energy consumption without negatively impacting quality of service (QoS) in the networks.

As they plan for the future, ICT site owners also need to consider the fundamental changes that are taking place in power generation and distribution. These changes, along with new energy consumption patterns, have created multiple challenges for the electricity utility sector. Mitigation alternatives will be costly and take a long time to deploy. By investing in a smart energy setup at ICT sites, ICT site owners can gain access to a resilient energy source that reduces their energy costs to net zero, while simultaneously opening up a promising new income stream by connecting with the smart grid.

Further reading

- » **Ericsson Technology Review, Building robust critical networks with the 5G system**, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/building-robust-critical-networks-with-the-5g-system>
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6. **Ericsson, Energy utilities**, available at: <https://www.ericsson.com/en/industries/energy-utilities>
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THE AUTHORS



Anette Höglund

◆ is the strategy execution driver for energy at the Ericsson CTO office. She has combined the perspectives of new business opportunities, product management and deployments in the field as well as development in different areas since she joined Ericsson in 1995. Höglund aligns Ericsson's technology strategy for energy solutions and evaluates the energy opportunity space with customer project collaborations. She holds an M.Sc. in management from Henley Business School, University of Reading, United Kingdom.

Johan Pettersson

◆ joined Ericsson in 1993 and currently serves as the strategic product manager for remote site management. For the past 10 years, he has mainly been working with



power and energy-related products linked to site equipment and the ability to monitor and reduce energy consumption at sites. He also drives Ericsson's business-wide energy program, whose purpose is to execute the technology strategy solution approved by the Ericsson Technology Board. Pettersson holds a B.Sc. in electrical engineering from Uppsala University, Sweden.



Helene Hallberg

◆ is a senior specialist in energy-efficient radio systems who joined Ericsson in 1988. She is

involved in energy-efficiency development and collaboration across the company, as well as in energy-related regulatory discussions and standardization activities. She is also a delegate in the International Electrotechnical Commission Technical Committee standardization of electricity supply systems encompassing transmission and distribution networks with their network interfaces. She has also filed a patent in the energy-efficiency area. Hallberg studied electrical engineering at a technical college in Stockholm, Sweden.



Erik Sanders

◆ is a product manager for AI and automation. He joined Ericsson in 2005 as an engineer in the 3G RAN area and continued with hardware development for radio base stations. In his current role, he drives innovation programs for Ericsson in the area of machine learning and reasoning. Sanders holds an M.Sc. in mobile communication from Linköping University, Sweden.



Lars Humla

◆ is a senior specialist in battery and solar systems at Ericsson. After working with battery solutions for mobile and fixed systems for the Swedish Defense Material Administration, Humla

joined Ericsson in 1996. He spent a brief period working for Emerson and as an agent for a battery technology company from Switzerland before rejoining Ericsson in 2006. He has been active in the Swedish standardization committee for batteries since 2010. Humla studied electrical engineering at a technical college in Stockholm, Sweden.

Energy-efficient packet processing in 5G mobile systems

Communication service providers around the world are looking for new opportunities to reduce energy consumption in their networks. As a complement to other approaches, we have evaluated a promising new method: applying micro-sleeps in packet processing nodes. Micro-sleeps can be opportunistically applied in all idle periods, thereby saving energy across a wide range of traffic conditions.

LEIF JOHANSSON,
PER HOLMBERG,
ROBERT SKOG

There are several ways to improve the energy efficiency of mobile communication systems and reduce the amount of energy it takes to provide mobile broadband services.

■ As part of its efforts to reduce network energy consumption, a communication service provider (CSP) may choose to upgrade to new hardware (HW) that consumes less energy than the previous generation. It is also possible to cut energy consumption with the help of an advanced scale-in/scale-out mechanism that reduces the amount of HW in use when traffic is low, such as during the night. In these conditions, some servers can be turned off, which results in energy savings. Regardless of the energy-saving mechanism,

however, it is essential to ensure there is no negative impact on real-time characteristics such as jitter and packet latency. It is also important to note that the correlation between Moore's law and power consumption savings for each new HW generation has decreased. Our latest research indicates that CSPs could save additional energy in their data centers by applying micro-sleeps in packet processing nodes. This approach has both lower overhead and lower latency than existing methods, and makes it possible to reduce power consumption even at high load, without degrading application performance. Because power management is implemented in communications libraries, it is easy to integrate into existing applications.

Packet processing in communication networks

Packet processing in communication networks involves the application of different functions and algorithms to ensure that each packet can run through the network efficiently. It includes steps such as packet identification, inspection and manipulation.

The network packet is a fundamental component in a packet-switched network. It consists of three main parts: the header, the payload and the trailer. The header contains information such as the sender address, the destination address, the length of the packet and the packet sequence number. The payload contains the transferred data – that is, one part of a video, e-mail or other type of content. The trailer marks the end of the packet and can also include error detection and correction information.

The different packet processing functions and algorithms range from fairly simple to more complex ones. A basic routing function is a good example of simple packet processing. More complex packet processing functions involve the application of different policies, charging and manipulation of the packets.

User plane packet processing nodes

While micro-sleeps can be applied to all types of packet processing, our research indicates that they are particularly impactful when used in the user plane function (UPF).

According to its definition in 3GPP, the UPF is a user plane packet processing node in mobile systems that links the RAN to the internet (or similar data networks) [4]. Simply put, the main purpose of the UPF is to forward packets to and from the internet. This is often done in combination with different

●● WHILE MICRO-SLEEPS CAN BE APPLIED TO ALL TYPES OF PACKET PROCESSING, THEY ARE PARTICULARLY IMPACTFUL WHEN USED IN THE UPF ●●

traffic optimization functions that range from pure Transmission Control Protocol (TCP) optimization to more advanced video optimization in combination with intelligent traffic congestion logic.

The UPF includes several complex processing functions such as GPRS Tunneling Protocol User Plane encapsulation and decapsulation. Other UPF functions include access control, bearer lookup, QoS mapping and marking. It also follows rules for guaranteed bitrate and maximum bitrate. Packets passing through the UPF are also subject to online/offline charging – that is, the application of different charging policies.

Instructions about how to process packets for different user equipment come from the session management function/policy control function. The processing of packets from different users occurs independently for the most part. All the packet processing functions and algorithms must meet the system requirement of real-time characteristics such as jitter and packet latency.

Kernel packet processing

The built-in, native networking support in computers is implemented as part of the operating system (OS) kernel and uses the POSIX (Portable Operating

Terms and abbreviations

AMD – Advanced Micro Devices | API – Application Programming Interface | CPU – Central Processing Unit | CSP – Communication Service Provider | DPDK – Data Plane Development Kit | HW – Hardware | NIC – Network Interface Card | OS – Operating System | RTD – Round-Trip Delay | SW – Software | UPF – User Plane Function

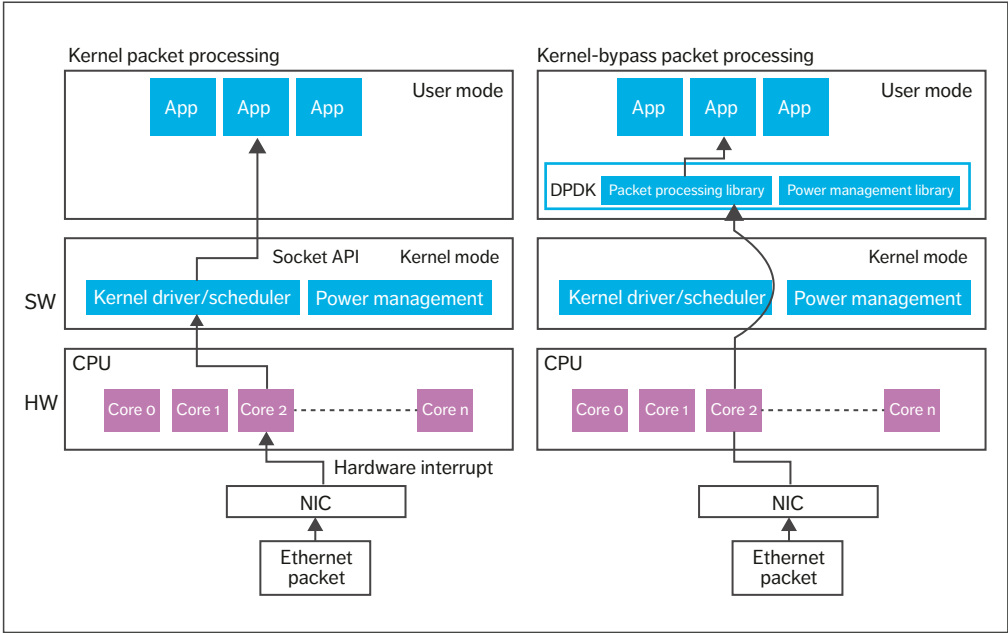


Figure 1 Kernel packet processing versus kernel-bypass packet processing

System Interface) socket as its standard application programming interface (API).

In kernel packet processing, user-mode application programs use the POSIX socket API to send and receive packets, while the kernel driver/scheduler handles the interaction with the network interface card (NIC). If the network is not ready (no packet has arrived) the kernel can decide to block that application while waiting.

The left side of Figure 1 illustrates how kernel packet processing works, with “App” representing user-mode applications. In this model, the OS is aware of the idle time, and OS power management can work to save power.

Kernel packet processing has multiple disadvantages. Most importantly, the high overhead of the OS calls and the copying of packets to/from the kernel space makes it hard to scale to high networking speeds and high packet rates. For example, the time between 1,534-byte packets on a

200Gbit Ethernet is 61ns. This corresponds to the execution time for performing a system call, which does not leave sufficient time to perform the packet processing itself. The speed of today’s network cards makes kernel-packet processing challenging at best, and impossible at worst.

Power management in the kernel can be problematic at high networking speeds. The OS kernel monitors and saves energy based on core utilization. This is a much slower mechanism that does not have the ability to follow the rapid changes in high-speed network traffic, resulting in queue buildups, delays and even packet drops.

Kernel-bypass packet processing

Kernel-bypass packet processing eliminates the kernel execution overhead by moving packet processing directly to the user-space, as shown on the right side of Figure 1. In this scenario, the OS can dedicate a network interface to an application such

as the Data Plane Development Kit (DPDK), which can program the HW from the user-space.

When DPDK is used, packets are received directly in user-space memory without kernel interaction, and all network-related interrupts are disabled. It is up to the application to make sure that packet queues and ring buffers to the NIC are checked frequently for new packets.

To avoid packet drops and reduce packet latency, the DPDK-based application is designed to check the packet ring buffer in busy-waiting mode, where the complete core is assigned to the application thread. This enables packet processing without any context switching or HW interruptions and minimal cache pollution, as the application thread is the only user of the core. Measurements using DPDK as a kernel bypass for packet processing show that millions of packets can be received in a single core and pipelined to other cores for further packet processing.

Kernel-bypass packet processing solves the issue of how to handle packet processing at speeds of 200 Gigabit Ethernet and beyond, but the busy-waiting technique of packet reception comes at a cost. No energy savings will be achieved if all the packet processing cores are 100 percent utilized in busy-waiting mode.

In kernel-bypass packet processing, Ethernet packets are transferred directly to the user-mode application memory. Power management needs to be performed in the user-space within the DPDK library. It is important to note that DPDK interacts with the kernel power management when changing core frequency [1].

Energy-efficient packet processing using DPDK

Higher data rates in packet processing have made it necessary to change packet processing to use DPDK with dedicated cores, user-mode execution and poll-mode drivers to remove the high overhead from context switches and system calls. Unfortunately, though, this solution is not as energy efficient as modern HW allows it to be. There are three main issues that need to be addressed:

- 1. The busy-waiting mode consumes more power than necessary when waiting for work.
- 2. OS power management is slow and cannot utilize idle time between packets and bursts of packets for high-speed interfaces.
- 3. The busy-waiting mode in DPDK always makes the core appear to be 100 percent utilized, with no information upon which to base power management decisions such as scaling down at periods of low traffic.

To solve these issues, power management actions must be fast, controlled directly from the packet processing application and executed in user mode. Two developments have made this possible. Firstly, server processors now have support for entering power-saving sleep states directly from applications in user mode. Secondly, the latest release of the DPDK library includes an option to utilize these sleep states while waiting for new networking events, making it possible to avoid the problems associated with the busy-waiting loop [1].

User-mode sleep states

Processors have specific instructions for entering sleep states, waiting for events and waking up. Previously these instructions were only available in the OS in supervisor mode, but new versions of these instructions have become available that are safe to use for implementing user-mode micro-sleep states. They are much faster than the old instructions and can be implemented without OS support.

The latest instructions allow a user-mode program to set up an event to monitor (that is, specify the update in the ring buffer that the program will

WHEN THE EVENT OCCURS (OR THE MAXIMUM WAIT TIME EXPIRES) THE PROGRAM SIMPLY WAKES UP AND CONTINUES EXECUTING

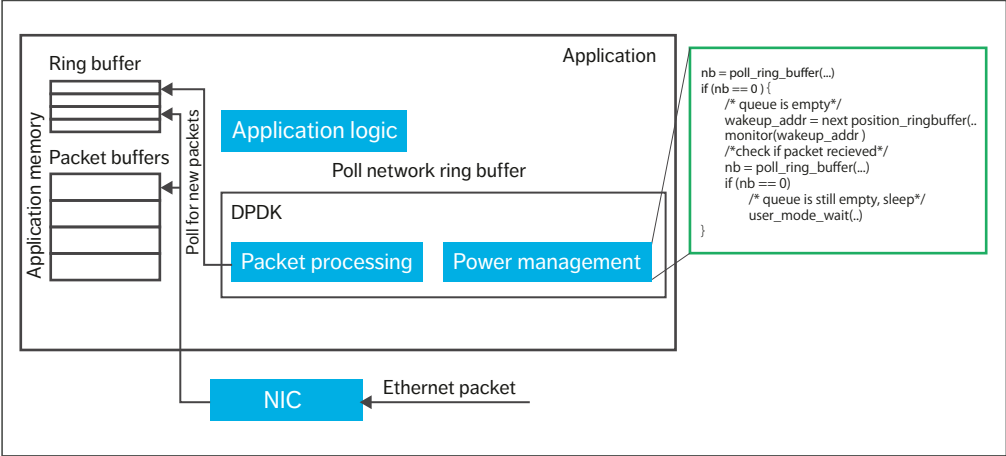


Figure 2 Power management integrated into the DPDK

wait for), define the maximum allowed wait time and then enter sleep state. When the event occurs (or the maximum wait time expires) the program simply wakes up and continues executing. There is no overhead to enter OS kernel, to do context switches and so on.

Examples of such user-mode instructions for setting up an event to monitor and for entering sleep state are UMONITOR and UMWAIT in Intel processors [2] and MONITORX and MWAIX in AMD processors [3]. These instructions make it possible to save energy for much shorter idle periods than was previously possible. They can also be efficient at higher communication speeds.

DPDK integration

Support for the new user-mode sleep-state mechanism has recently been added to the DPDK library, which makes it possible to enter energy-saving sleep states directly in the poll loop doing busy-waiting of DPDK descriptor ring or rings.

If the polls of the descriptor ring (or rings) are empty, an event monitor is set up and a final poll is done before user-mode sleep state begins. Relevant updates – such as an NIC or other central processing unit (CPU) core updates related to the monitored position in the ring – will be detected by processor

HW, which will then wake up the CPU core to restart execution of the next instruction, continuing the poll. The poll loop will then detect the update.

Figure 2 illustrates how power management is integrated into the DPDK. In this setup, the network device allocates the ring buffers and packet buffers in the application user-mode memory and updates the ring buffer with the new packet buffer point at packet reception. User-mode sleep state is entered when the ring buffer is empty (that is, the queue to the NIC is empty). Wake-up is triggered when the NIC updates the ring buffer. It is important to note that the proposed power management does not impact the application logic.

Observability

Processors include a measurement functionality for time spent executing and time spent in different sleep states. User-mode sleep states make the CPU thread utilization visible for this measurement functionality, and the CPU thread utilization can also be observed for the DPDK application. CPU thread utilization can also serve as input to the control loops for other types of power management (such as frequency scaling) that may be used as a complement to user-mode sleep states, potentially leading to further improvements in energy efficiency.

Hardware offload

HW offload is a technique that transfers the handling of selected flows from the processor (the slow path) to the NIC (the fast path). This approach frees up the CPU core and makes micro-sleep technology more effective because a larger number of CPU cores have the opportunity to micro-sleep. HW offload also improves the efficiency of the Peripheral Component Interconnect bus and offers other potential benefits with regard to improving throughput, efficiency and latency in the system.

Processor core voltage/frequency scaling

Scaling down CPU core voltage and frequency is a complementary method for achieving the best efficiency over longer periods of lower processing. There is a trade-off with user-mode sleep state, as scaling down also means shorter idle times.

Since power consumption increases proportionally to the square of the voltage, it is favorable to scale down the voltage and frequency when the packet rate drops for a period of time. A fast and reliable monitoring function is needed to detect increased packet rates and, most importantly, packet bursts. The faster and more reliable the detection is, the more aggressively the voltage and frequency can be scaled down without causing negative effects and degrading KPI due to queue buildups that increase latencies, timing jitter and the associated risk of packet drops.

DPDK provides interfaces that enable fast control of CPU core voltage/frequency scaling. In our implementation, we use a fast packet-burst detection in DPDK together with a control loop that maintains a performance margin that handles even the worst-case burst scenarios. When there is a margin, the user-mode wait states offer an alternative method of energy saving, and we can avoid any negative impact on packet processing without compromising energy efficiency.

Processor uncore voltage/frequency scaling

Uncore refers to the parts of the processor outside the actual cores – that is, the interconnection between the CPU cores (and other functions) and

THE RESULTS OF THESE EXPERIMENTS INDICATE THAT THE POWER-SAVING METHODS ARE STABLE AND EFFICIENT

the interfaces on the processor that are shared and used by all CPUs. Scaling down the voltage and frequency of the uncore is a complement to energy-saving mechanisms implemented in the core. As in the core, it is favorable to scale down the voltage and frequency in the uncore when the packet rate drops for a period of time.

As the uncore is shared by all CPU cores, frequency needs must be coordinated between all cores and applications, including those outside DPDK, making uncore changes a slower mechanism than core frequency changes.

Experimental validation and results

We ran two sets of lab experiments to validate the user-plane micro-sleep design. In the first set, we generated a workload that we consider to be realistic and applied it to a UPF application. The UPF measurements showed that processor energy consumption decreased from 190W to 61W at low traffic load and from 190W to 145W at maximum traffic load. Beyond quantifying the potential energy savings in a realistic use case, our results verified that power-saving methods work well with existing communication applications.

Our second set of experiments used synthetic workloads that were designed to simulate extreme conditions. The results of these experiments indicate that the power-saving methods are stable and efficient, with negligible impact even in worst-case conditions.

Energy-efficient packet processing applied on a user plane function

In the first set of experiments, we applied the DPDK-based power management with micro-sleep and

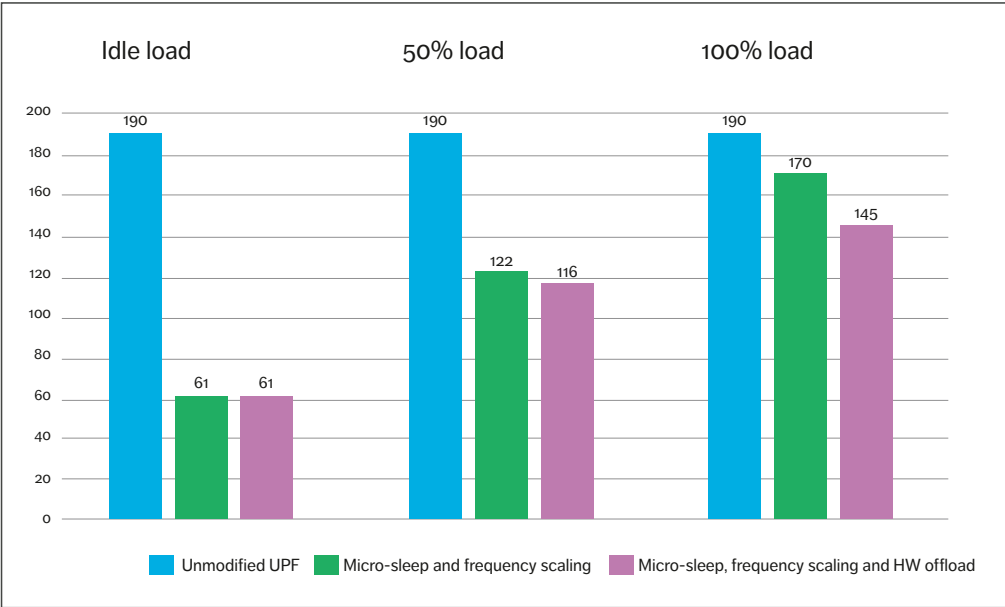


Figure 3 CPU power consumption at idle load, at 50 percent maximum traffic load and at 100 percent maximum traffic load

frequency scaling on an Ericsson UPF application. To measure the energy gain when using HW offload, the UPF is designed to off-load packet flows with the highest bandwidth to the network device.

As a traffic model, we used a traffic mix from our internal stability test. We generated the traffic using an external traffic generator that simulates the surrounding network elements. Figure 3 presents the results. The blue bar shows CPU power consumption when running the DPDK-based UPF unmodified at idle load, at 50 percent of the maximum traffic load and at 100 percent traffic load. The green and purple bars show the energy consumption when running the same traffic load with power management without HW offload, and with both power management and HW offload.

The measurements show a 68 percent processor power reduction at idle load when using power management (the green bar on the far left in Figure 3). The bars for 50 percent load and 100 percent load with enabled power management show the

reduction in processor power even as traffic load increases.

It is difficult to perfectly balance the processor load and thereby enable the cores that are not fully loaded to sleep and save energy. In our case, 10 percent of the energy could be saved at maximum traffic load (the green bar on the far right in Figure 3). This result will vary depending on the application and traffic mix.

NIC HW offload technology enables further opportunities for micro-sleep, leading to an overall 24 percent reduction at maximum load (the purple bar on the far right in Figure 3). The amount of power that can be saved using HW offload will depend on the ability of each application to utilize it and thereby save processor resources.

Synthetic benchmarks

Our proposed power management solution using micro-sleep and frequency scaling targets packet processing with an extremely low impact on packet

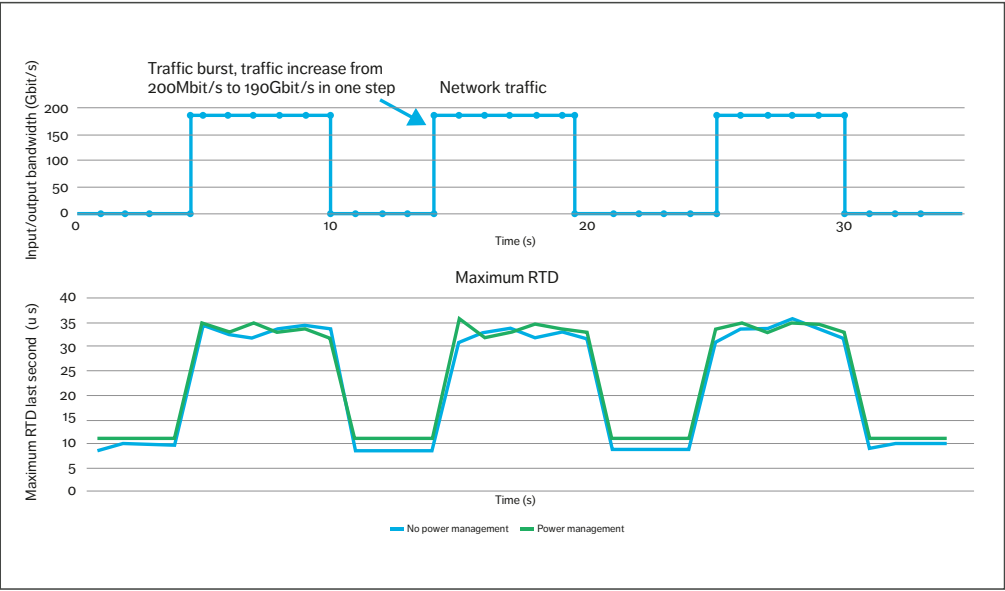


Figure 4 Measured maximum RTD when applying extreme traffic bursts, showing no increased delay when using power management

latency and a fast reaction to traffic bursts. For example, in low-latency power management, traffic bursts must not cause any packet drops. The new user-mode sleep states enable the core to wake up within a few 100ns and should have minimal impact on the total packet latency.

To measure the packet latency and the speed at which the power management functionality can react to traffic bursts, we used a synthetic benchmark where the application's impact on the packet latency is reduced as much as possible when receiving network traffic. This enabled us to measure the packet latency impact when using user-mode sleep states and frequency scaling. The synthetic benchmark is also used to measure how fast the system can scale up the performance when receiving traffic bursts.

The synthetic benchmark program fetches Ethernet packets using DPDK, swaps the media access control address and sends the packet back to the original source. The source machine measures

the time from the sending of the packet to the receiving of the packet (known as the round-trip delay (RTD)) including packet drops. User-mode sleep is used when queues to the network device are empty and frequency scaling is triggered when network traffic increases/decreases.

Figure 4 shows the maximum measured RTD down to the last second, measured on the external traffic generator. At low traffic, the RTD is slightly higher when using power management (the green line in the bottom half of Figure 4) due to reduced core/uncore frequency. When traffic increased from 27Mbit/s to 190Gbit/s, the core/uncore frequency changed to maximum speed using the implemented DPDK burst detector. The power management impact of the packet delay is negligible at maximum load (190Gbit/s).

Our results indicate that the measured worst-case RTD or maximum packet jitter is not impacted, even if the RTD value is slightly higher at low traffic load. No packets were dropped during the measurements.

Conclusion

Our research clearly demonstrates that is possible to reduce energy consumption while simultaneously meeting the requirements of latency-sensitive applications. The latest generation of server processors enables fast power-saving transitions by entering sleep states directly from applications in user mode, with no impact on important KPIs like jitter and packet latency. With no overhead to enter the OS kernel and no context switches, the central processing unit core simply wakes up and continues executing when packets arrive during user-mode sleep state. Because power management is part of the user-mode packet processing library, energy

savings are easy to integrate and validate on existing Data Plane Development Kit-based applications. The combination of micro-sleep, hardware offload and frequency scaling has the potential to lead to significant energy savings. Our experiments on an Ericsson user plane function show that processor energy consumption decreased from 190W to 61W at low traffic and from 190W to 145W at maximum traffic. We also carried out experiments using a synthetic benchmark, which demonstrate that our proposed power management solution has no measurable impact on worst-case round-trip delay and worst-case packet latency.

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Further reading

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THE AUTHORS



Leif Johansson

◆ is an expert in the characteristics of open systems. He joined Ericsson in 1996 after receiving an M.Sc. in physics from Uppsala University, Sweden. Johansson has more than 20 years of experience evaluating and applying new technology in Ericsson's product portfolio.



Per Holmberg

◆ is a computer architect. He holds an M.Sc. in electrical engineering from KTH Royal Institute of Technology in Stockholm, Sweden, and has worked at Ericsson since 1985 designing processing solutions primarily for communication equipment. Holmberg has held



specialist and expert positions in computer architecture and processing systems, been lead architect for several processor and computer designs and is responsible for more than 50 inventions.

Robert Skog

◆ is a senior expert in the field of service architecture. After earning an M.Sc. in electrical engineering from KTH Royal Institute of Technology in 1989, he joined Ericsson's two-year trainee program for system engineers. Since then, he has mainly worked on end-to-end solutions and traffic

optimization for everything from the first WAP solutions to today's advanced user plane solutions. In 2005, Skog won Ericsson's prestigious Inventor of the Year award.

Improving energy performance in 5G networks and beyond

Continued focus on energy performance in 5G and 6G development will be essential to enable new deployment scenarios with smaller and lighter telecom equipment, as well as minimizing the climate impact of mobile networks.

CECILIA ANDERSSON,
JONAS BENGTSSON,
GREGER BYSTRÖM,
PÅL FRENGER,
YLVA JADING,
MY NORDENSTRÖM

As the 5G rollout continues and we look ahead to 6G, the exponential growth in processing is going to be one of the biggest challenges from an energy performance perspective.

■ The telecom industry has a long history of prioritizing peak performance and high capacity. Until recently low energy consumption was typically perceived as a nice-to-have extra benefit rather than a crucial feature of state-of-the-art mobile network equipment. As awareness about the need to optimize energy performance in telecom networks continues to grow, several challenges must be addressed.

In 5G networks, digital processing in base stations can increase more than 300 times compared with early Long Term Evolution (LTE) products, primarily due to an increasing number of antenna branches, broader bandwidths and shorter transmission time intervals (TTIs). This increase is expected to become even larger in 6G. To handle this increase responsibly from an energy consumption perspective, it is essential that the future 6G standard is as lean as possible. Most importantly, the amount of mandatory and always-on signaling must be kept to a minimum, which will enable underlying components and subsystems to provide sufficiently high levels of load dependence in their energy consumption.

The percentage of energy consumption originating from compute and digital silicon will continue to rise as the New Radio (NR) rollout continues. The trends of increased bandwidth, more antenna ports and shorter TTIs are driving an exponential increase in processing needs on the digital frontend, which propagates into the radio unit, beamforming and layer one processing. Processing needs for layer two, packet processing and control functions are also increasing, but not as quickly.

What is energy performance?

In existing mobile systems, where energy consumption has low dependence on the load and traffic growth is high, an energy-efficiency metric is not particularly useful. Even if no effort is made to reduce energy consumption, traffic growth will cause energy efficiency to improve from one year to the next.

The term energy performance broadens the focus beyond energy per bit to total energy consumption, highlighting the similarities between achieving high system performance and low energy consumption. Optimizing energy performance means minimizing the energy consumption for a set of performance requirements (user throughput, capacity, latency and so on). The concept of energy performance makes the trade-off between energy consumption and performance requirements transparent.

Energy performance is relevant in three separate contexts – economy, ecology and engineering – that have different stakeholders and use different terminology. In the economic context, optimizing

THE TERM ENERGY PERFORMANCE BROADENS THE FOCUS BEYOND ENERGY PER BIT TO TOTAL ENERGY CONSUMPTION

energy performance results not only in lower operating expenses for mobile network operators, but also in the potential for lower capital expenses due to the use of smaller and lighter equipment. This equipment enables new and simplified deployments as well as less-costly power supply and energy storage solutions. Low energy consumption is therefore of particular importance when discussing the costs for enhanced availability and enhanced national security.

In the ecological context, the value of optimizing energy performance is best demonstrated by the fact that about 94 percent of Ericsson's total carbon emissions impact is associated with the operation of our products [1].

The engineering context is arguably the most underestimated and underutilized aspect of energy performance. A sharp focus on reduced energy consumption in the product engineering phase will yield desirable reductions in product size and weight. Tighter energy performance product requirements drive technical, managerial and organizational innovation in a greater change process that leads to closer collaboration over multiple areas of technology.

Terms and abbreviations

AI – Artificial Intelligence | **C-RAN** – Centralized RAN | **CSI** – Channel-State Information | **D-RAN** – Distributed RAN | **KPI** – Key Performance Indicator | **LTE** – Long Term Evolution | **MB** – Megabyte | **Mbps** – Megabits Per Second | **MHz** – Megahertz | **MIMO** – Multiple-Input, Multiple-Output | **ms** – Millisecond | **NR** – New Radio | **PA** – Power Amplifier | **PRB** – Physical Resource Block | **RAN** – Radio Access Network | **RAT** – Radio Access Technology | **RF** – Radio Frequency | **Tbit** – Terabit | **TTI** – Transmission Time Interval | **UE** – User Equipment | **W** – Watt

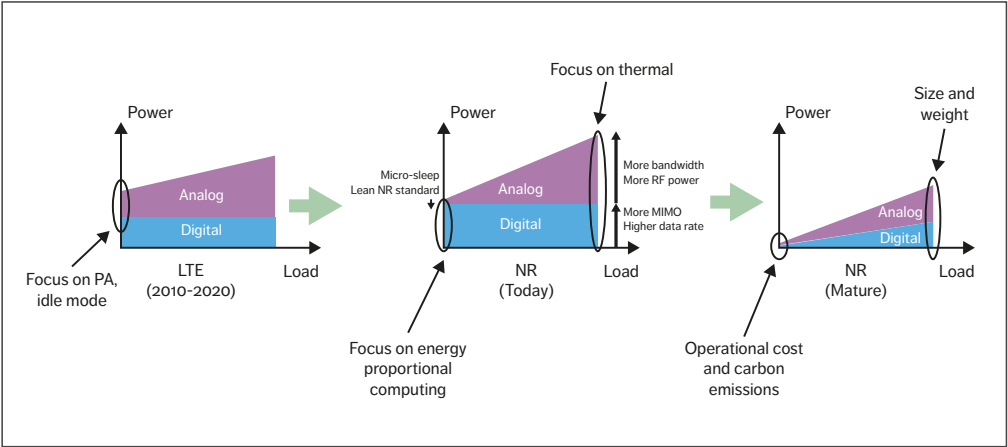


Figure 1 The energy performance journey of mobile networks

Depending on the context, the impact of energy usage can be expressed in terms of money, carbon emissions, thermal design constraints and so on. But regardless of how the benefit is viewed, the solutions required to reduce energy consumption are very similar. It is important, however, to differentiate between targets whose purpose is to reduce average energy use (mostly relevant for operational costs, carbon emissions and the like) and those aimed at reducing peak energy use (mostly relevant for product dimensioning, thermal design, size and weight). Fortunately, though, effective solutions for energy reduction often reduce peak and average energy use simultaneously.

The network energy performance journey

Figure 1 illustrates the energy performance journey of mobile networks. The graph on the left shows the power consumption of a typical LTE base station in the period 2010-2020. The graph in the middle shows the power consumption of a typical New Radio (NR) base station today. The graph on the right shows the projected power consumption of a mature NR base station beyond 2025.

In LTE, the energy consumption of the radio access network (RAN) was dominated by base

stations that comprised around 80 percent of the RAN electricity use. Furthermore, within each base station, around 80 percent of the energy consumption was used in the power amplifiers (PAs) [2]. During this time the main energy performance goal was to reduce the idle-mode consumption of the PAs.

Network traffic was much lower in the early days of LTE. In a countrywide network, only 5 percent of the physical resource blocks (PRBs) were typically used for data traffic, while around 95 percent of the PRBs were empty. The high static energy consumption and low average traffic resulted in very low load dependency in the network, where the additional energy increase due to the traffic in the network was well below 2 percent of the total energy used [3].

In order to improve the load dependency, we introduced micro-sleep transmission, an energy-saving feature that deactivates and reactivates PAs in microseconds. Micro-sleep transmission is effective at all times when there are no transmissions from the base station. However, it could be even more effective if the LTE standard was not filled with mandatory and always-on reference signals that cannot be turned off, even when there is no traffic. The obvious solution to this was to redesign

OUR RESEARCH SHOWS THAT IT IS POSSIBLE TO MOVE FROM TODAY'S SITUATION TOWARD A MORE ENERGY-LEAN TOMORROW

the physical layer so that only a minimum of signaling would be necessary when there is no data to transmit, but this required a new standard.

When 5G NR was developed, we ensured that its physical layer had an ultra-lean design [4]. This design makes features such as micro-sleep transmission much more efficient, and the effect on live networks is already significant [5]. Micro-sleep transmission has had a major impact over the course of the past decade by dramatically decreasing the energy consumption in the analog radio parts of base stations.

5G NR is also designed for massive MIMO (multiple-input multiple-output), and it supports wider bandwidths than 4G LTE. A typical LTE base station has two transmit and receive branches, 20MHz of spectrum, and the digital processing time in the base station is 1ms (corresponding to one TTI). Early NR products have 64 antenna branches, support 100MHz of spectrum and have a TTI of 0.5ms. This implies that digital processing in radio base stations needs to increase 320 times compared with early LTE products.

In today's NR products, the energy that digital components use can be as large, or larger, than the power used by the analog components (mainly PAs). The additional bandwidth and additional antennas were also introduced while keeping the radio frequency (RF) power spectral density constant at around 2W/MHz. While a typical 20MHz LTE base station could deliver 40W RF output power, a 100MHz NR base station can be capable of delivering 320W RF power.

More bandwidth and more RF power increased the peak power usage of analog components, while more antenna branches and more digital processing

increased the power consumption of digital components (see the middle part of Figure 1). This massive upscaling of capabilities has resulted in an increasing energy consumption trend in the industry. Fortunately, it is possible to alter this trend with new building practices. Our research shows that it is possible to move from today's situation (the middle part of Figure 1) toward a more energy-lean tomorrow (the right part of Figure 1) by taking the following actions:

- 1. Improve the load dependence of components (especially digital ones) by significantly improving sleep modes in terms of idle-mode energy usage, reactivation times and granularity. One example of a technique that is currently underutilized in many digital designs is dynamic voltage and frequency scaling (DVFS).
- 2. Relax the requirements for analog components dynamically (when possible).
- 3. Implement more power management functions in the network nodes (requires more load-dependent components to be effective).
- 4. Use artificial intelligence (AI) techniques to achieve network-level optimizations (requires more load-dependent network nodes to be effective).
- 5. Consider removing support for legacy technologies such as WCDMA (Wideband Code Division Multiple Access) and GSM (Global System for Mobile Communications).

This last point is particularly important, as supporting legacy technologies results in requirement creep and consumes development and testing resources, thereby reducing the resources available for the implementation of power-saving functions.

Successful execution of these actions would not only optimize energy performance; it would also enable operators to reduce total network energy consumption by adding additional capacity as traffic increases, as more capacity results in more idle mode operation [6].

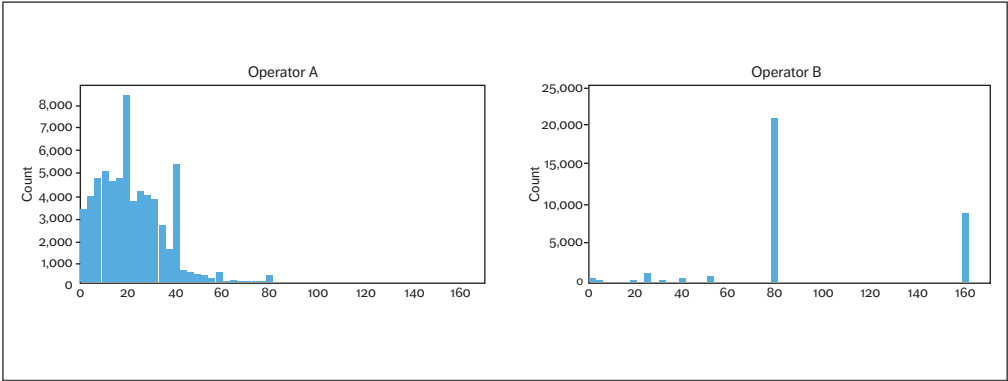


Figure 2 Histograms of configured maximum transmission power for two European operators

Insights from field data

While some operators run their networks with the power settings of every base station fixed at the maximum level, others have more optimized and diverse configurations. Figure 2 provides histograms of configured maximum transmission power for two European operators. Operator A, on the left, optimized the power levels in its network, while Operator B, on the right, used default maximum power settings.

One study on this topic [7] indicates that the total network energy usage can be reduced by around 10 percent without compromising performance simply by tuning the output-power levels, as shown in Figure 2. There are similar differences in how existing energy-saving features such as MIMO sleep mode, booster carrier sleep, cell deep sleep and the low energy scheduling solution are activated (or not) in different networks. This indicates a clear need for improved tools to optimize the energy performance in existing networks.

Meaningful reference points

Some energy-saving solutions are associated with a negative key performance indicator (KPI) impact. However, we would argue that this often depends on the reference KPI chosen to compare with, or perhaps even more importantly, what target KPI is expected to be fulfilled.

Figure 3 illustrates the total downlink traffic volume (purple) and the mean user throughput (blue) in a real network for one day. Due to variations in traffic volume, the user throughput differs by about a factor of two between the peak traffic hour and the minimum traffic hours. The question is, when evaluating the throughput obtained with an energy-saving feature turned on, should we relate that to the performance during peak hour or during the minimum traffic hour? The large KPI variations that we normally observe in the network are not always desirable, and an energy-saving feature might even be an attractive way to both obtain energy and cost savings, as well as more predictable throughputs in a network throughout the day.

It is not necessary to set the desirable KPI for an energy-saving feature to the minimum KPI obtained at peak hour. It can also be set to other targeted KPIs for a specific network, such as a KPI for which the network has been dimensioned. The network dimensioning KPI target can be regional and/or service dependent. In some areas such as city centers or indoor factories, the requirements may need to be higher, while in other areas they can be more relaxed. In the end, this should be an operator policy decision. One could, for example, allow a fraction of the performance above the minimum performance requirement to be used to reduce energy consumption.

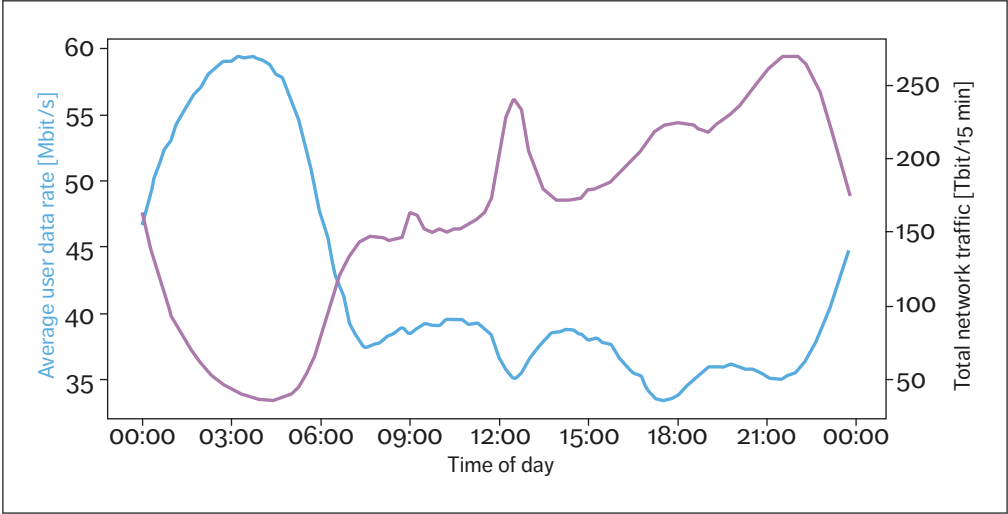


Figure 3 Variations in traffic volume (purple) and user throughput (blue) over a day in a real network

Centralized versus distributed radio access networks

It is often argued that a centralized RAN (C-RAN) is inherently better than a distributed RAN (D-RAN) deployment for enabling low-network power-consumption solutions. As mobile networks are dimensioned for peak traffic, resources are always overprovisioned at some places in the network. As peak traffic occurs at different times of day in different locations of a network, a C-RAN can exploit pooling gains when processing resources can be shared in a common pool. In addition to the processing-pooling gain, C-RAN deployments can also utilize more efficient cooling, power-supply and energy-storage solutions. Based on this, considerable gains have been reported [8].

It is important to understand which parts of a network that can realistically be centralized in a C-RAN deployment and under what circumstances. C-RAN deployments are typically assumed to operate on off-the-shelf hardware, which opens up the possibility to add value with custom-made silicon that can deliver higher load dependence and lower fixed energy costs. Increased latency is a

consideration that will limit the centralization of the more time-critical and power-consuming lower-layer processing, such as the digital frontend. The analog (radio) parts must also remain distributed in a C-RAN deployment, as the users will still be distributed.

Traffic: RAN for mobile broadband and beyond

There are two main approaches to reducing energy usage in RAN: rush to sleep, and rate adaptation. The rush-to-sleep approach aims to transmit the data as fast as possible in order to maximize the time in sleep mode. Rate adaptation instead aims to adapt the transmission rate to instantaneous requirements and thereby enable energy savings by under-clocking or deactivating some components while transmitting data.

When looking at traffic statistics in real networks, it is evident that both of these approaches are needed. About 95 percent of all data sessions are small (less than 1MB), and the one percent largest sessions contribute to almost three quarters of the total data volume. For small data sessions, rate adaptation is an effective way of reducing the energy

needs, as large bandwidth and many antenna ports are not required for most sessions. In contrast, the best way to handle large sessions from an energy performance perspective is by combining high data rates with effective mechanisms for the equipment to rush to sleep once the transmission is finalized.

Video traffic is currently estimated to account for 69 percent of all mobile data traffic, a share that is forecast to increase to 79 percent in 2027 [9]. To deliver a satisfactory user experience, a network must be able to play video content on demand, without delays or stalling. Video streaming uses buffering to smooth out throughput variability. As a result, video is a service that can easily be made compatible with power-saving features in the RAN.

The ability to avoid stalling requires rate adaptation on a relatively slow timescale (seconds). Ensuring a sufficiently short time-to-play for video services and time-to-content for web services requires an almost instantaneously available bitrate in the order of 20Mbps [10]. As acceptable time-to-content and time-to-play numbers can be in the order of one to four seconds, “almost instantaneously” needs to be significantly smaller than this (100ms, for example).

For web and video services, it is therefore sufficient for a base station to provide about 20Mbps to any user within 100ms if additional capacity to accommodate the new service can be made available within one second. This would result in around 1.5 seconds time-to-content, which is classified as excellent in the 2025 scenario in a recent Ericsson Mobility Report [9].

Critical machine-type communication and other low-latency services can also be made compatible with RAN-power saving features if the low-latency requirements are managed with care. Low-latency service requirements must be broken down into two additional categories: service-registration latency and protocol ramp-up latency. The activation of a low-latency service requires a quality of service guarantee from the RAN, which includes a setup procedure that can be allowed to take 100ms or more.

In addition, low-latency requirements need to be

contained regionally and per band. It is not necessary to provide countrywide support for ultra-low latency to enable factory automation in very limited areas. All the bands, cells and nodes that support low-latency services are not needed all the time. Even the most critical low-latency service does not require a network that is constantly in a high-alert state that prohibits the use of energy-saving rate adaptation or capacity adaptation in the RAN.

In short, it is possible to support all services today and in the future (as far as we know) while deactivating close to 100 percent of the excess capabilities in a base station if the following two conditions are met:

1. At least 20Mbps can be provided to an additional user within 100ms.
2. Any additional required capacity and rate can be made available within one second.

Any active but unutilized spare capacity beyond this is a waste of energy.

Standardization and 6G

System design by standardization is the foundation that enables energy-efficient design of an entire network. But having a good standard is not enough – components and products must utilize the potential that the standard provides. Network management that includes the application of AI tools for load balancing and energy consumption minimization is also essential. To be effective, network management requires a good standard that enables low-energy and load-dependent operation, as well as products that utilize the energy savings that the standard enables.

NR is a lean standard that is already a powerful enabler of low network-energy usage. The most important thing from an energy performance perspective moving toward 6G is to maintain and extend the lean properties on which NR is based, such as enabling up to 160ms of transmission-free periods. For 6G the concept of lean design should be extended to better support network densification and even larger massive MIMO antenna arrays

without requiring excessive spatial repetition and beam sweeping of idle-mode signals regarding synchronization, system information and paging.

Today’s NR specifications provide signaling support for idle-mode user equipment (UE) to see the full set of beams, bands and nodes that are configured and that can be made available in active mode. The need for this transparency in standards and products is questionable, given the energy cost of associated transmissions and shorter transmission-free periods, as well as reduced sleep possibilities.

In the discussions about next-generation technologies, there is a tendency to allow requirements from one area to propagate into other areas, or even into all areas. The support for extreme capabilities – such as extremely high data rates with corresponding extreme transmission bandwidths, extremely low and predictable latency and extreme reliability – is important to enable the wide range of use cases envisioned for 6G.

At the same time, such capabilities come with a cost in terms of network energy consumption. It is crucial that this cost is limited to the situations where and when the specific capabilities are required. One way of doing this is to prevent requirements for active mode from also applying to idle mode. We would argue that it is worth considering a stricter separation between active and idle mode in future networks.

In addition, efforts should be made to design functionality so that it is self-contained, refraining from the reuse of signals specified for one functionality to support other. This may sound counterintuitive, but experience shows that the associated dependencies between different functionalities often prevent desirable sleep-mode possibilities. A prime example of this is the cell-specific reference signals in LTE that are used for both active-mode demodulation of data and for idle mode UE cell search and mobility. This reuse results in a very high cost for transmitting signals in idle mode.

One of the improvements we would like to see in 6G is the ability to avoid the overhead cost of

obtaining channel-state information (CSI) when there is no data to transmit. A more preamble-based design of the physical radio links, where synchronization signals and reference signals for CSI acquisition are transmitted together with the data bursts, would make the cost of all supporting signals (synchronization, CSI acquisition and so on) explicit. More opportunistic scheduling of such supporting signals would become more natural as a result.

The availability of more shared and pooled infrastructure can contribute to further lowering total network energy consumption. Examples of this include multi-radio-access technology (RAT), multi-operator and/or multi-band operation. Although there already are several standardized solutions for operations in such scenarios, further enhancements are worth investigating.

With regard to the role of AI in energy performance improvements, the vast majority of AI functionality is expected to be related to implementation rather than specification. However, standardized AI-supporting functionality related to observability and control that targets energy optimization at network level would be helpful.

As the work on 6G progresses, it is important to be mindful of the risks associated with introducing new functionality into later releases of technology specifications. The potential of lean design can easily be compromised both in standards and implementation, depending on how new functionality and the associated signaling is added.

To summarize, 5G is already a very good standard that enables the implementation of low energy-consuming behavior in mobile systems. Lean design can be further optimized and enhanced in 6G by including additional domains (frequency bands, beams, nodes, RATs, slices and so on).

Conclusion

Lean design is the foundation for improving radio access network energy performance. The lean design of New Radio (NR) standard was a major improvement compared with Long Term Evolution (LTE), enabling unprecedentedly low energy consumption in live 5G networks. It is of the utmost

importance that this progress is not compromised in future standardization and products.

Looking ahead, the main energy performance challenge will be scaling processing with traffic to meet the digital processing needs of high-performing networks with larger antenna arrays and bandwidths, and shorter processing requirements. In the 6G timeframe, we are advocates for a further reduction in fixed idle-mode energy usage and peak power requirements – changes that will enable the use of smaller, lighter products and support novel deployment solutions in 6G networks.

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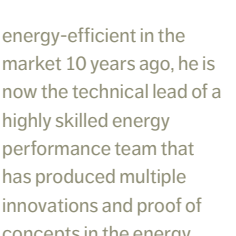
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THE AUTHORS



Cecilia Andersson

◆ joined Ericsson in 2007 and currently works as a system designer specializing in RAN energy performance. She is a leading promotor of integrating and incorporating the network aspects of energy performance to reduce the total energy consumption of RANs. Andersson holds a Ph.D. in physics from Uppsala University, Sweden.



Greger Byström

◆ is a section manager for radio product development, with extensive experience working with functional systemization of several of the energy performance functions within Ericsson radio products. Before joining Ericsson in 2011, he studied engineering physics at Umeå University, Sweden.



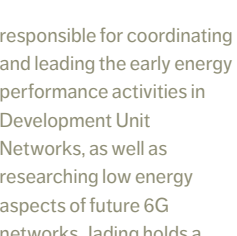
Jonas Bengtsson

◆ is a principal developer specialized in RAN energy performance. One of the drivers behind making Ericsson modems the most



Ylva Jading

◆ joined Ericsson in 1999. She currently works as a senior specialist in network energy performance at Ericsson Research. She is

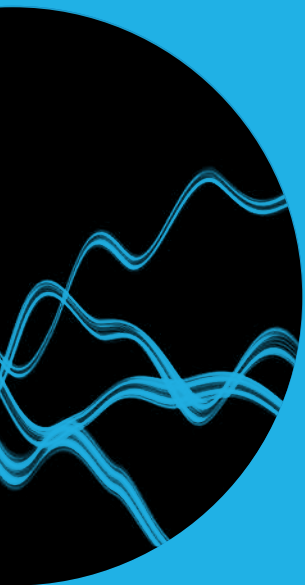


My Nordenström

◆ is a system developer within Business Area Networks. She joined Ericsson in 2013, and in her current role she primarily focuses on reducing the energy consumption and improving the energy performance for Ericsson's RAN compute products. Nordenström holds an M.Sc. in design and product realization from KTH Royal Institute of Technology, Stockholm, Sweden.

Further reading

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Ericsson
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Phone: +46 10 719 0000